

THE METHOD 1.6 THE INDEX OF INDICES

Matthew Lach

The Method 1.6 — a reading copy of the seated volume,
pressed 5 September 2026

Contents

I · THE ATOMIC INDEX ITSELF	5
II · THE TOWER	6
III · THE INDEXES DRAWN BESIDE IT	7
IV · THE VIOLATION INDEX	8
What the companion prints	8
Its defect, and the one constraint printed	8
What is not printed, and what nine searches established	9
V · THE INDEXES BUILT FROM IT	10
The electromagnetic quotient	10
The time index	10
The space index	11
Λ_3 — the three-body index	11
The Kreuzer–Skarke frontier	12
The string partition function	12
The languages	12
The book’s own indexes	13
How much of an index determines the rest	13
The channel survey — $\Lambda_{\text{spectra}}^{\text{obs}}$	14
VI · THE INDEXES NAMED AND NOT BUILT	17
VII · THE FULL TABLES	18
Λ_8 — all 976 cells	19
The periodic table — all 90 cells, (period, group)	42
Janet — all 118, $(n+\ell, Z)$	42
The calendar — 365 cells, (month, day)	43
A box ordering — all 35, $l \geq w \geq h$	43
VIII · WHAT EACH INDEX IS, AND WHAT ONLY IT CONTRIBUTES	44
Λ — the atomic index	44
The tower $\Lambda_8 \dots \Lambda_{13}$	44
The periodic table — (period, group)	44
Janet’s left-step table — $(n+\ell, Z)$	44
The nuclide chart — (Z, N)	45
The calendar — (month, day)	45
The box ordering and the chessboard	45
The electromagnetic quotient	45
Λ_{spectra} — the channel index	46
The violation index	46
Λ_{phys} — the parameter index	46
The languages	47
The book’s own indexes	47
Λ_3 — the three-body index	47
The one-line summary	48

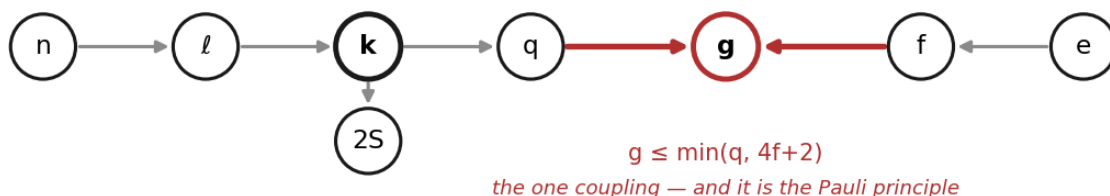
IX · THE INDEXES BUILT IN THE LÖWDIN WORK	49
Λ_{law} — the laws of the channel equation.....	49
Λ_{const} — the constants	49
Λ_{var} — the variables.....	49
Λ_{ryd} — the Rydberg series	49
Λ_{charge} — the roles of the charge	50
Λ_{cross} — the crossing values	50
Λ_{descent} — the charge dependence.....	50
Λ_{phys} — the parameters, rebuilt.....	50
Λ_{amp} — the electron–electron term	51
Λ_{PCA} — physics $\oplus$ charge $\oplus$ amplitude, merged	51
Λ_{chem} — the chemical properties	51
Λ_{t} — the corridor position	52
Λ_{ladder} — the ladders themselves	52
Λ_{xray} — the inner shell	53
The one-line summary, extended.....	54
X · THE INDEXES BUILT IN THE LÖWDIN SOLUTION	55
Λ_{chain} — the derived filling index	55
Λ_{cinf} — the counterfactual twin	55
Λ_{V5} — the contested-row closure	56
Λ_{j120} — the unwitnessed extension	56
XI · TRANSITIONS, THE COMPLETE TABLE.....	57

Compiled 2026-08-15. **Every index this work builds on or beside the atomic index — what it holds, whether it closes, whether it carries time, and what role it plays for Λ .** Measured where measurable; where an index is named and never built, that is stated rather than filled in.

I · THE ATOMIC INDEX ITSELF

$\Lambda = \{ (n, \ell, k, q, e, f, g, 2S) \in \mathbb{N}^8 : \text{eight constraints, at caps} \}$.

Λ_8 · eight coordinates · seven constraints · a caterpillar with 2S pendant at k



A cell is a **transition**: a source subshell (n, ℓ) holding k electrons, q of them moving, into a target subshell (e, f) that ends with g , at spin $2S$. **Not a state — a move.** Everything else in this document follows from that.

constraint	reads
$\ell \leq n - 1$	hydrogenic — the subshell fits its shell
$k \leq 4\ell + 2$	Pauli — the subshell's capacity
$q \leq k$	conservation of the removed — you cannot take more than is held
$f \leq e - 1$	hydrogenic again, on the target
$g \leq 4f + 2$	Pauli again, on the target
$g \leq q$	conservation of the placed
$2S \leq k$	the multiplicity a subshell can carry
$k \geq 1$	<i>definitional</i> — a transition needs a mover

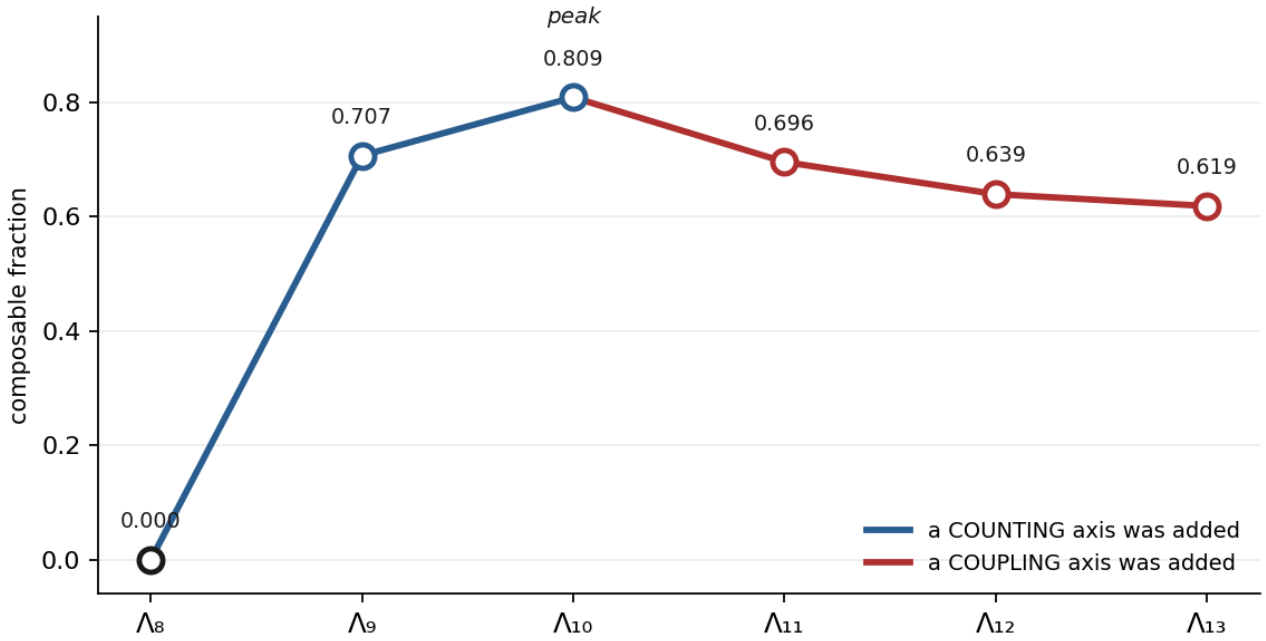
The one coupling is $g \leq \min(q, 4f+2)$, the single non-product term in the generating function, and it is the Pauli principle. Remove it and 976 becomes 1,000; the 24 excluded cells all have $f = 0$ and $g = 3$ — three electrons in an s orbital.

II · THE TOWER

Each stage adds one axis and inherits every bound below.

stage	cells	axes	composable	fraction	the axis added
Λ_8	976	8	0	0.0000	—
Λ_9	1,654	9	1,169	0.7068	$2S' \leq g$
Λ_{10}	2,535	10	2,050	0.8087	$2S' \leq v \leq g$
Λ_{11}	13,585	11	9,450	0.6956	$2J_c \leq \varphi(k)$
Λ_{12}	70,905	12	46,740	0.6592	$2K \leq 2J_c + 2f_{\max}$
Λ_{13}	199,130	13	127,070	0.6381	$ 2J - 2K \leq 1$

Composability up the tower — counting axes raise it, coupling axes lower it



Counting axes raise the composable fraction; coupling axes lower it. No exception.

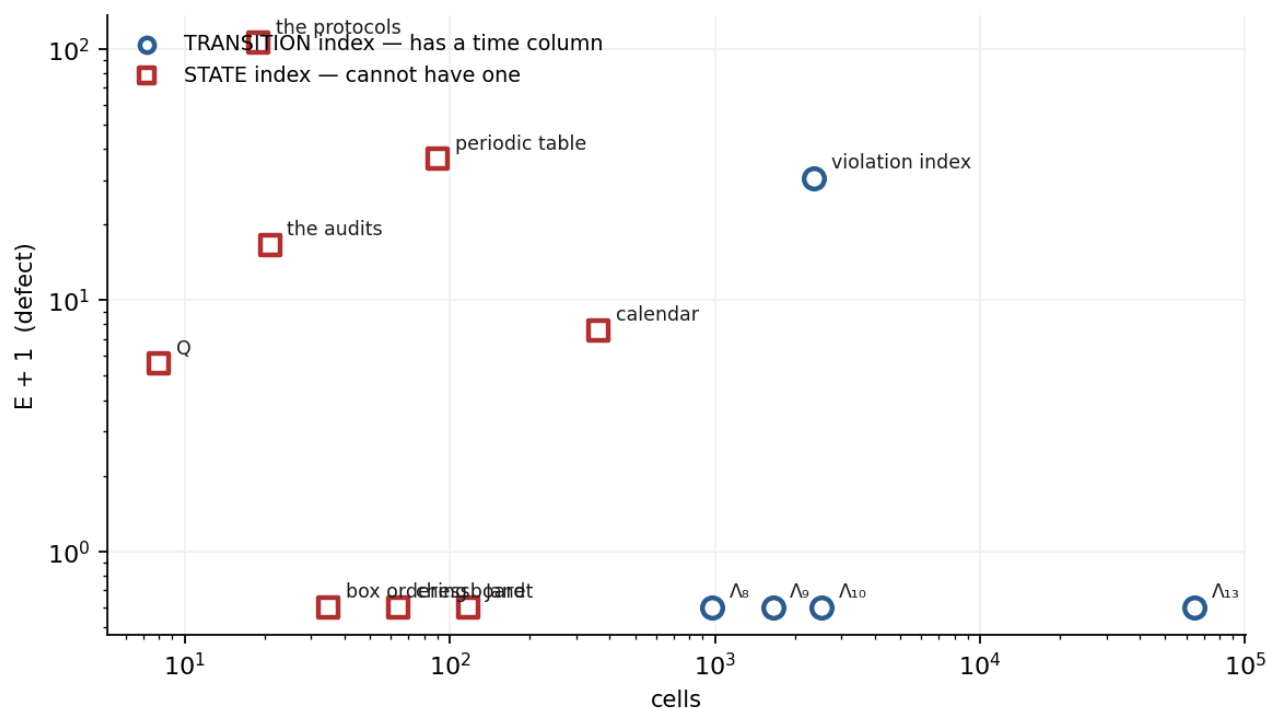
Λ_8 composes not at all — four source coordinates against three target. Λ_{10} is the peak at 80.87%, and every stage after it falls. The non-composable cells are exactly those with $g = 0$ at the peak, since $k \geq 1$ forbids an empty target being a source.

III · THE INDEXES DRAWN BESIDE IT

index	coordinates	cells	box	E	kind
the periodic table	(period, group)	90	126	36	state
Janet's left-step	($n+l$, Z)	118	944	0	state
the calendar	(month, day)	365	372	7	state
a box ordering	(l , w , h)	35	125	0	state
a chessboard	(rank, file)	64	64	0	state

Reading the table. $E = 0$ carries information only when the ambient box exceeds the cells, which is why box is reported beside every defect. **E is the closure defect, not the difference between the two columns** — Janet's left-step and the box ordering both close at $E = 0$ inside boxes far larger than their cell counts, while the chessboard closes at $E = 0$ because its box IS its cells.

The index landscape — every index this work builds or draws



None of these is a transition index, so none can carry a time column. A state cell has one position; the question *is my target another cell's source* does not arise.

Janet closes at $E = 0$ and the classroom table does not. That is a fact about the two tables **in coordinates chemists fixed for other reasons** — §21.6 shows any composite count has SOME coordinate system where $E = 0$, so the result is that Janet's 1928 ordering, chosen for shell filling, happens to close.

IV · THE VIOLATION INDEX

The companion paper's object, indexed over nine letters. It is here because it is the one index this work reasons about without holding — **its cells are not printed anywhere**, and what follows is everything that is.

letter	meaning	rungs
X	exotic matter required	4
S	semiclassical corrections	3
IC	initial conditions	3
U	unitarity	3
NEC	null energy condition points	5
L	locality	2
SD	superdeterminism	2
DNc	dynamical, continuous	3
DNd	dynamical, discrete	3

$4 \cdot 3 \cdot 3 \cdot 3 \cdot 5 \cdot 2 \cdot 2 \cdot 3 \cdot 3 = 19,440$, and the index holds **2,370** — 12.19% density.

What the companion prints

object	threshold	cells	reachable
classical black hole	none	2,370	1,410
Hawking-evaporating	$NEC \geq 1$	2,196	1,410
Planck-scale wormhole	$NEC \geq 2$	1,764	1,134
macroscopic wormhole	$NEC \geq 3$	1,146	738
universal horizon	$X \geq 2$	1,374	840
time machine	$X = 3$	558	360

Twelve numbers, and one column is a second predicate on the same cells. Differenced, the cells give $NEC = 0$ at 174, $NEC = 1$ at 432, $NEC = 2$ at 618, $NEC \geq 3$ at 1,146. **All 174 cells at $NEC = 0$ are unreachable** — removing them removes no reachable cell — and **exactly 64% of $NEC = 1$ and 64% of $NEC = 2$ cells are reachable**, 276 of 432 and 396 of 618.

Its defect, and the one constraint printed

E = 30, collapsing as 1×30 — one core cell repeated across the free coordinates — with the core at (**X = 0, U = 0, NEC = 3**). At fifteen letters: 18,072 cells, **E = 816 = 1×816** . **Multiplicities 3, 5, 10, 30 at 4, 5, 7 and 9 coordinates**, and 5 is not a product of any two rung counts, so **the edge list constrains the free letters as well as the support**.

The constraint the companion prints, at arity 3:

| $NEC \geq 3 \wedge X = 0 \rightarrow U \geq 1$ — cells 2,370, E = 30, core 1

with three variants: $NEC \geq 3 \rightarrow U \geq 1$ at arity 2 giving **E = 0**, and $NEC \geq 3 \wedge U = 0 \rightarrow X \geq 1$ and $NEC \geq 3 \rightarrow (IC \vee U \vee X)$ both giving **E = 30**. **A jurisdicted forcing and an unjurisdicted disjunction give identical defects; arity 2 gives none.**

What is not printed, and what nine searches established

The edge list itself. §5.7 of the companion says so outright. Nine searches were run against the twelve numbers; the best holds **cells exactly 2,370 at distance 29**, with the whole residual on the NEC interior — **NEC ≥ 2 sixteen too high, NEC ≥ 3 nine too low, missing in opposite directions.**

Eight alphabets were tried and none moved it: simple implications, disjunctive heads, conjunctive bodies, all 47 single-rule edits, four alternative rung readings, two threshold semantics, seeding at NEC = 3, and weighted sums with gated deviations. **The rungs and the $\geq$ reading are confirmed correct** — every alternative is two orders of magnitude worse.

And the reachable column reduces to the same missing artefact. Three local readings of reachability give ~99% against a printed 60%; the companion's own here? column says reachable means reachable FROM HERE, directed from one position in a transition graph. **Both columns need the edge list.**

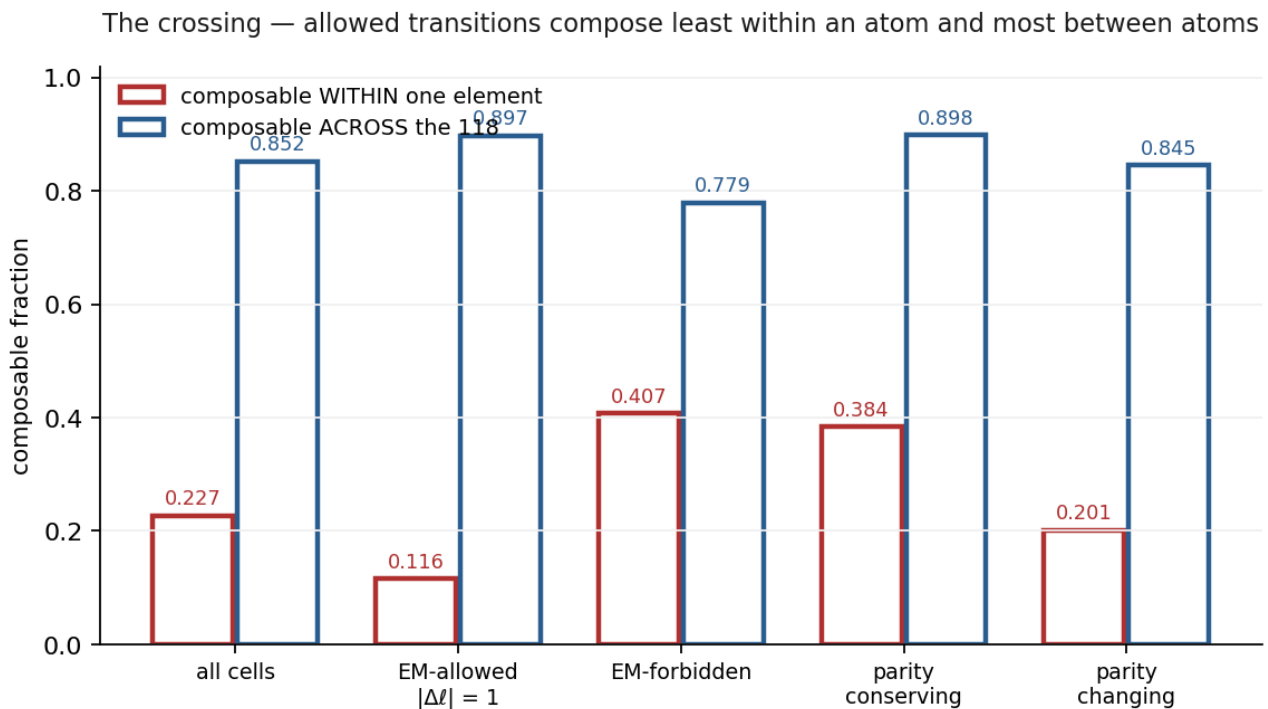
V · THE INDEXES BUILT FROM IT

Each of these exists because Λ exists — it indexes something Λ 's cells have, or something the act of building Λ produced.

The electromagnetic quotient

Λ modulo the dipole selection rules — $\Delta\ell = \pm 1$, $\Delta S = 0$, parity. A quotient, not a subset: cells are identified when the rules cannot distinguish them.

Its $E = 0$ is VACUOUS — §12.11.8 — because the image is a complete rectangle. **It closes because it is a box, not because the selection rules constrain it.**



Its measured role is the crossing. Within one element, EM-allowed transitions compose at **11.6%** against the forbidden **40.7%**. Across the 118 elements the order reverses — **89.7% against 77.9%**. *The rule that forbids composition inside an atom is the rule that enables it between atoms.*

The time index

Not a coordinate. The second column. §12.11.0: composition IS the temporal order — a cell's source end is a before, its target end an after, and q is what changed between them.

An index has a time column exactly when its cells are moves. Λ_8 976/0, Λ_9 1,654/1,169, Λ_{10} 2,535/2,050 — and the periodic table and the calendar **cannot have one**, because a state cell has one position and there is nothing to compose.

What Λ lacks is not a clock but a denominator — claims per session — which §12.11.1.1 now supplies. *A date must never enter Λ : a transition is a type, and types are not dated.*

The space index

The first column. What the index holds, as against what it can compose. **Λ has no spatial coordinate:** n and e are shell numbers, not positions, and nothing in the eight is a place.

The nearest thing to a spatial reading is the observability one: an observer's register is Λ -valued because observation IS an atomic state change, so an unregistered event is an admitted-and-absent cell and **$E(\text{local register})$ measures what is not yet real to that observer.** *That is a definition with a falsifiable consequence and it is not yet built.*

Λ_3 — the three-body index

The second index built from physics rather than from Λ (the first is Λ_{spectra}), and the first whose subject is classical. Chapter 36; register 1713–1724.

coordinate	values	bound
stratum	KAM, per, chaos, erg, coll	five, exhaustive
E	$\mathbb{R}$	sign fixes bounded/unbounded
L	$\mathbb{R}$	—
masses	$\mathbb{R}_+^3$	13 order-types; symmetry order 6, 2, 1

Fig. 3 The tower read downward: each dropped axis coarsens the product (§12.11, §17.4).

axes dropped	index	product
12 inputs (p_1, p_2, p_1, p_2)	full phase space $\mathcal{M}_{\{E, L\}}$ ↓	trajectory $r_i(s)$
4 inputs (w_1, w_2, p_1, p_2)	planar reduced S^2 geodesic flow ↓	reduced path $w(s)$
2 inputs (E , braid word)	periodic stratum $\mathcal{M}_{\text{per}}$ ↓	closed loop in B_3
1 input $\mathbb{C} = L^2 E /G^2M^5$	ergodic stratum $\mathcal{M}_{\text{erg}}$	distribution $P(\varepsilon)$

$E(\Lambda_3) = 0$. Certificate (§18.4.1): the shape map, three dropped coordinates. *Its cells are configurations, not moves, so it carries no time column — the flow is the geodesic flow of the Jacobi–Maupertuis metric, and time is a quadrature (§12.11.1.3).* Named at §12.11.2 as the maximal case of what Chapter 18 forbids; built, and the completeness is the impossibility theorem read as an index. ## The nucleon index

The measured nuclide chart, indexed by proton and neutron count. Chemistry's index is Λ ; the nucleus has its own, and it is the one place a second physical index is built alongside the atomic one. Register 298, 388, 390.

coordinate	values	bound
Z	$\mathbb{Z}_+$	proton number
N	$\mathbb{Z}_+$	neutron number

$E = 9$, on the particle-bound nuclides at low Z — those that do not immediately emit a nucleon — stable across four proton-number cutoffs. Closing that chart admits nine cells it lacks: **He-5, He-7, Li-10, Be-8, Be-13, B-9, B-16, B-18, C-21**, each a known unbound nuclide interior to the chart. Be-8 is unbound because it is two alpha particles — clustering; He-5, He-7, Li-10 and Be-13 are unbound by one neutron against an even-paired core — pairing. **The nine are the pairing and clustering terms of the mass formula, counted** — content the coordinates cannot carry, since (Z, N) records proton and neutron count and not whether four nucleons form an alpha. The defect appears by $Z \leq 7$ and none is added after, so it is a property of the measurement, not of the cutoff. The drawn chart between its drip lines is $\mathcal{R}$ of the measured one, $E = 0$ by idempotence.

The Kreuzer–Skarke frontier

The Calabi–Yau catalogue, indexed by its two Hodge numbers. An index the method reaches outside atomic physics — the reflexive-polytope list of string compactifications, drawn beside Λ to show the operator travels. §31.3.

coordinate	values	bound
h^{11}	$\mathbb{Z}_+$	first Hodge number
h^{21}	$\mathbb{Z}_+$	second Hodge number

$E = 540$ on a specifiable slice. The full catalogue is a file this work could not read, but one slice is stated exactly in print: the $\chi = \pm 6$ points, with Hodge numbers $(h, h+3)$ and $(h+3, h)$ for $13 \leq h \leq 128$ and named exclusions — **208 cells**. Closing it admits **540 cells**, from 498 join failures and 498 meet failures of 21,528 pairs. The 540 are predictions, not a defect: the join of $(h, h+3)$ and $(h+3, h)$ is $(h+3, h+3)$, so $\mathcal{R}$ fills the diagonal the two Hodge-pair sequences bound. The admitted cells carry five distinct Euler characteristics, $\chi \in \{0, \pm 2, \pm 4\}$, with 112 on the diagonal from $(13, 13)$ to $(131, 131)$ and $h^{11} + h^{21}$ from 26 to 262. Each is a Hodge pair the catalogue must contain if the pair set is closed under Batyrev’s duality; every one satisfies the published bounds $h^{11} \geq 1$, $h^{21} \geq 1$ and $h^{11} + h^{21} \leq 502$, so none is refuted, but the file to confirm them is missing. The index proposes; it is not thereby right.

The string partition function

The generating function of string state degeneracies, read as an index with multiplicity. Where Λ ’s cells are present or absent, this index’s cells carry a count — the number of string states at each level. §31.2.3.

coordinate	values	bound
mode occupation	$\mathbb{Z}_+$ per oscillator (n, i) , $n \geq 1$, $i \in 1 \dots 24$	independent
level	$N = \sum n \cdot \text{occ}(n, i)$	the graded total

$E = 0$, because the modes are independent. The partition function $\prod_n (1 - q^n)^{-24} = \sum d(N) q^N$ counts the states at level N as the 24-coloured partitions of N — a part of size n available in each of the 24 transverse dimensions — giving $d(1) = 24$, $d(2) = 324$, $d(3) = 3,200$, and $d(14) = 156,883,829,400$. The admissible set of occupation vectors is a full product, each oscillator’s occupancy free of every other’s, and a product closes trivially: $\mathcal{R}$ adds nothing, so $E = 0$. It closes for the reason dual to Λ ’s — Λ ’s coordinates couple along a tree and close in one pass; the string’s modes do not couple at all, which is the degenerate case of the three closure conditions where independence stands in for the tree. The two are the same theorem at its two ends: an index closes when its coordinates are free, and it closes when they couple only along a tree, and nothing between.

The languages

A language is a coordinate system; translation is re-coordinatisation; E is the cost.

language	closure operator	delete	add
order	$\mathcal{R}$	restored	absorbed
geometry	monotone polyhedron	restored	absorbed
algebra / logic	ideal, Gröbner basis	still derivable	T·h enters the ideal
analysis	coefficients of F	restored	absorbed
information	description length	E_bits stays 0	grows
statistics	max-entropy on the marginals	restored	grows
documentary	none — no algorithm	—	—

Six agree on Λ at 976 and all ten pairs hold — C(5,2), a complete graph, not a ladder. **Statistics was tested and qualifies with a caveat:** it recovers Λ at 976 and $E = 0$, but gives $E = 0$ on the periodic table where $\mathcal{R}$ gives 36 — *marginals cannot see a hole*. It agrees with the others only where there is nothing to disagree about.

The book's own indexes

index	coordinates	cells	E	what it holds
the audits	ob- ject<source<artefact<outs ide · what · how · depends	21	16	twenty-two audits
the register	corroboration $\leq \phi(\text{repair})$	33	6	the corrections
Q	blocks · obstacle · cost · depends	8	5	what the book does not know
the protocols	trigger · object · failure · earned	19	105	twenty-four protocols
the constraints	parents · form · carried · source · role	15	21 / E₄ 6	every constraint in the book
the mathematics	kind · language · status · verification · precedent	77	0	fibred, twenty-four fibres (27 over sixteen when this table was first printed; main §D.5.10)
the numbers	fibre · kind · role	—	40	every figure the book prints
the channel survey	Z · core charge · ℓ	1,664	1,351	596 Rydberg channels across 313 cells

These exist because the book exists, not because Λ does — but they are built by Λ 's method, and the audits index's two frontier cells are the same pair $\mathcal{R}_4$ refuses at §14.5.6, reached by two computations sharing no code.

How much of an index determines the rest

Remove cells at random and ask whether $\mathcal{R}$ puts them back. The largest fraction removable with exact recovery is a property of the index, and it had never been measured for any of these.

index	dimension	envelopes	coupling	redundancy
Λ	8	56	29%	61%
$\Lambda_{\text{spectra}}^{\text{obs}}$	3	6	17%	20%
a box ordering	3	6	50%	0%
Janet	2	2	50%	0%
the periodic table	2	2	0%	0%
the calendar	2	2	0%	0%

Coupling does not explain it. The fraction of coordinate pairs whose envelope actually constrains gives $\mathbf{r}^2 = 0.002$, $\mathbf{p} = 0.94$ against redundancy: Janet and the box ordering both couple at 50% and recover nothing, while $\Lambda_{\text{spectra}}^{\text{obs}}$ couples at 17% and recovers a fifth. *That conjecture was stated before the measurement and is wrong.*

DIMENSION explains it, and the projection test shows so on a single object. Λ restricted to its own first d coordinates, constraints unchanged:

d = 8	61%	d = 5	30%
d = 7	30%	d = 4	5%
d = 6	30%	d = 3	0%

$\mathcal{R}$ works on **pairwise** envelopes, so an index of dimension d carries $d(d-1)$ of them — **56 for Λ , 6 for $\Lambda_{\text{spectra}}^{\text{obs}}$, 2 for a plane**. With two envelopes there is almost nothing to reconstruct from, and that is what the operator is rather than a fact about any subject.

But only INDEPENDENT coordinates count. Adding the electron count $N_e = Z - c + 1$ to $\Lambda_{\text{spectra}}^{\text{obs}}$ — a function of two coordinates it already has — takes redundancy from **20% to 0%**, while doubling the envelopes and raising coupling from 17% to 33%. *A derived coordinate adds no information and obliges $\mathcal{R}$ to reproduce it exactly, so recovery becomes strictly harder.* Registers 1119–1122.

The channel survey — $\Lambda_{\text{spectra}}^{\text{obs}}$

Coordinates: $Z \cdot \text{core charge} \cdot \ell$. Each cell is a Rydberg channel — a fixed parent core, a fixed ℓ , n running. The compendium holds **596 channels across 313 of the survey’s 1,664 cells**.

This is not the coordinate index, and the two were once both called Λ_{spectra} . Λ_{spectra} proper is the four-coordinate index (Z , charge, ℓ , $2S+1$) at 104,832 cells with $\mathbf{E} = \mathbf{0}$ — see Λ_{spectra} — *the channel index*, below. $\Lambda_{\text{spectra}}^{\text{obs}}$ is the three-coordinate **survey of what has been measured**. The column beside each cap below is grid minus held — the **unwitnessed** count — and *not* a closure defect: reading it as E would say the index is open when it is a fixed point. R 1657.

Where its alphabets come from, and why that differs from Λ . Λ ’s coordinates are bounded by the physics — $\ell < n$, $k \leq 4\ell+2$ — and all 118 ground configurations are known within its caps, so Λ is complete at 976 cells. **$\Lambda_{\text{spectra}}^{\text{obs}}$ is not.** Rydberg series are unbounded in n and ℓ runs to $n-1$, so the index is **infinite unless capped**, and every cap is a decision. Three defensible ones give:

alphabet	grid	unwitnessed
as measured — charge [1, 2, 3, 4, 5, 6, 9, 11, 15, 16]	1,664	1,351
charge 1 to $Z-1$ over the elements held	4,376	4,148
all 118 elements, ℓ 0–7	55,224	54,996

All three grids are closed — each is a fixed point of $\mathcal{R}$, verified by double projection. The number beside each is what has *not* been measured under that cap, so it is a result about the cap and about the state of the literature; it is never a closure defect. *Registers 959–962, 1657.*

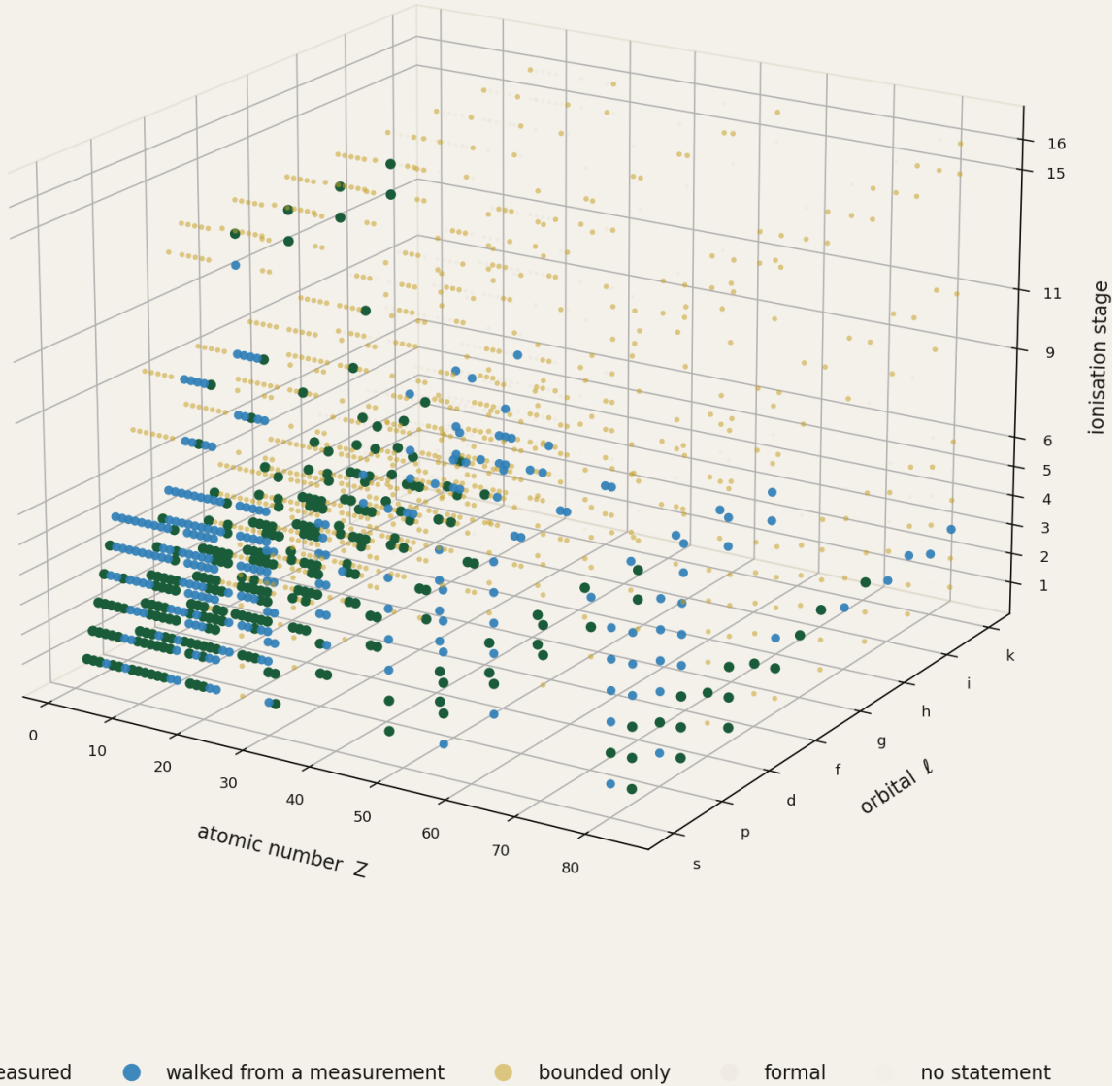
What the index says about cells it does not hold. Every cell carries a grade, and the grade records how far the statement travelled:

grade	cells	what it means
MEASURED	205	a defect computed from levels
BRACKETED	5	bounded above and below by DIFFERENT mechanisms
BOUNDED	42	bounded on one side, adjacent to a measurement
PROPAGATED	954	a bound inherited through a chain of 2 to 13 steps
FORMAL	73	the only bound is one a mechanism already implies
UNCONSTRAINED	33	no statement at all

97.5% of the index carries a defensible statement. The three relations that propagate are **the l-collapse (register 693)** along ℓ , **the isoelectronic ladder (register 708)** along an isoelectronic sequence, and **the same-element ladder (register 963)** along the charge states of one element. *Registers 963–965, 981.*

Λ_{spectra} — the index, and how much of it carries a number

1,664 cells · 313 measured · 243 walked · the green wedge is what has been captured, the blue what follows from it



Walking reaches s and p at high electron count, and $\ell \geq 4$ through Seaton. The gap at $\ell = 2, 3$ is where no step exists.

And the last 33 are of three kinds, only two of which a capture can fix. Ar I holds one cell at d — an ordinary gap. **P I, Cd I and Ti II** hold six each, eighteen cells at $\ell \geq 2$, wanting a longer capture. **Sc I and Ti I have no Rydberg series at all** — their levels are $3d.4s.4p$ and $3d^2.4s$ mixtures with no n running, so their **fourteen** cells are unreachable by any capture. One, eighteen and fourteen make the 33. *Registers 983, 984. The ceiling is 98.9%, not 100% — $(1,312 - 14) / 1,312$ — and that is a fact about the open-shell transition metals rather than about the collection.*

VI · THE INDEXES NAMED AND NOT BUILT

§29.1: the absence is not evidence. These were named in the course of the work and no index exists for them. Saying so is the point of this section.

Sound. Phonon modes are discrete and indexable — ω , lattice momentum, polarisation, occupation — but they index a **material**, not an atom. **No such index is built here**, and it would sit beside Λ rather than inside it, since Λ has no coordinate a lattice vibration could occupy.

Frequency. A transition's frequency is $\Delta E/h$, and ΔE is not a coordinate of Λ — **§6.2 gives first ionisation energies to the bracket and the bracket REFUSES**, at $\text{Be} \rightarrow \text{B}$, $\text{N} \rightarrow \text{O}$, $\text{Mg} \rightarrow \text{Al}$ and $\text{P} \rightarrow \text{S}$, four steps of twelve. *That refusal is the book's most direct statement about frequency: the coordinates cannot bracket it, and where they refuse a mechanism is entering.*

The multiverse. §32.6.1 places this outside all four of the book's indexes, with the destination coordinate, because §E.1.4 shows **the open set cannot express an unclosable question**. *It is not an index that has not been built. It is the shape of a question this work cannot pose* — and Theorem 18.1 is why: Λ 's order structure contributes nothing predictively beyond what the coordinates supply, and its coordinates are $n, \ell, k, q, e, f, g, 2S$.

VII · THE FULL TABLES

Every index small enough to print, printed. The tower above Λ_8 runs to 199,130 cells at Λ_{13} and is given by its construction instead; the violation index has no cells to give.

Λ_8 — all 976 cells

Λ_8 — all 976 cells

#	n	ℓ	k	q	e	f	g	2S
1	1	0	1	0	1	0	0	0
2	1	0	1	0	1	0	0	1
3	1	0	1	0	2	0	0	0
4	1	0	1	0	2	0	0	1
5	1	0	1	0	2	1	0	0
6	1	0	1	0	2	1	0	1
7	1	0	1	0	3	0	0	0
8	1	0	1	0	3	0	0	1
9	1	0	1	0	3	1	0	0
10	1	0	1	0	3	1	0	1
11	1	0	1	1	1	0	0	0
12	1	0	1	1	1	0	0	1
13	1	0	1	1	1	0	1	0
14	1	0	1	1	1	0	1	1
15	1	0	1	1	2	0	0	0
16	1	0	1	1	2	0	0	1
17	1	0	1	1	2	0	1	0
18	1	0	1	1	2	0	1	1
19	1	0	1	1	2	1	0	0
20	1	0	1	1	2	1	0	1
21	1	0	1	1	2	1	1	0
22	1	0	1	1	2	1	1	1
23	1	0	1	1	3	0	0	0
24	1	0	1	1	3	0	0	1
25	1	0	1	1	3	0	1	0
26	1	0	1	1	3	0	1	1
27	1	0	1	1	3	1	0	0
28	1	0	1	1	3	1	0	1
29	1	0	1	1	3	1	1	0
30	1	0	1	1	3	1	1	1
31	1	0	2	0	1	0	0	0
32	1	0	2	0	1	0	0	1
33	1	0	2	0	1	0	0	2
34	1	0	2	0	2	0	0	0
35	1	0	2	0	2	0	0	1
36	1	0	2	0	2	0	0	2
37	1	0	2	0	2	1	0	0
38	1	0	2	0	2	1	0	1
39	1	0	2	0	2	1	0	2
40	1	0	2	0	3	0	0	0
41	1	0	2	0	3	0	0	1
42	1	0	2	0	3	0	0	2
43	1	0	2	0	3	1	0	0
44	1	0	2	0	3	1	0	1

#	n	ℓ	k	q	e	f	g	2S
45	1	0	2	0	3	1	0	2
46	1	0	2	1	1	0	0	0
47	1	0	2	1	1	0	0	1
48	1	0	2	1	1	0	0	2
49	1	0	2	1	1	0	1	0
50	1	0	2	1	1	0	1	1
51	1	0	2	1	1	0	1	2
52	1	0	2	1	2	0	0	0
53	1	0	2	1	2	0	0	1
54	1	0	2	1	2	0	0	2
55	1	0	2	1	2	0	1	0
56	1	0	2	1	2	0	1	1
57	1	0	2	1	2	0	1	2
58	1	0	2	1	2	1	0	0
59	1	0	2	1	2	1	0	1
60	1	0	2	1	2	1	0	2
61	1	0	2	1	2	1	1	0
62	1	0	2	1	2	1	1	1
63	1	0	2	1	2	1	1	2
64	1	0	2	1	3	0	0	0
65	1	0	2	1	3	0	0	1
66	1	0	2	1	3	0	0	2
67	1	0	2	1	3	0	1	0
68	1	0	2	1	3	0	1	1
69	1	0	2	1	3	0	1	2
70	1	0	2	1	3	1	0	0
71	1	0	2	1	3	1	0	1
72	1	0	2	1	3	1	0	2
73	1	0	2	1	3	1	1	0
74	1	0	2	1	3	1	1	1
75	1	0	2	1	3	1	1	2
76	1	0	2	2	1	0	0	0
77	1	0	2	2	1	0	0	1
78	1	0	2	2	1	0	0	2
79	1	0	2	2	1	0	1	0
80	1	0	2	2	1	0	1	1
81	1	0	2	2	1	0	1	2
82	1	0	2	2	1	0	2	0
83	1	0	2	2	1	0	2	1
84	1	0	2	2	1	0	2	2
85	1	0	2	2	2	0	0	0
86	1	0	2	2	2	0	0	1
87	1	0	2	2	2	0	0	2
88	1	0	2	2	2	0	1	0

#	n	ℓ	k	q	e	f	g	2S
89	1	0	2	2	2	0	1	1
90	1	0	2	2	2	0	1	2
91	1	0	2	2	2	0	2	0
92	1	0	2	2	2	0	2	1
93	1	0	2	2	2	0	2	2
94	1	0	2	2	2	1	0	0
95	1	0	2	2	2	1	0	1
96	1	0	2	2	2	1	0	2
97	1	0	2	2	2	1	1	0
98	1	0	2	2	2	1	1	1
99	1	0	2	2	2	1	1	2
100	1	0	2	2	2	1	2	0
101	1	0	2	2	2	1	2	1
102	1	0	2	2	2	1	2	2
103	1	0	2	2	3	0	0	0
104	1	0	2	2	3	0	0	1
105	1	0	2	2	3	0	0	2
106	1	0	2	2	3	0	1	0
107	1	0	2	2	3	0	1	1
108	1	0	2	2	3	0	1	2
109	1	0	2	2	3	0	2	0
110	1	0	2	2	3	0	2	1
111	1	0	2	2	3	0	2	2
112	1	0	2	2	3	1	0	0
113	1	0	2	2	3	1	0	1
114	1	0	2	2	3	1	0	2
115	1	0	2	2	3	1	1	0
116	1	0	2	2	3	1	1	1
117	1	0	2	2	3	1	1	2
118	1	0	2	2	3	1	2	0
119	1	0	2	2	3	1	2	1
120	1	0	2	2	3	1	2	2
121	2	0	1	0	1	0	0	0
122	2	0	1	0	1	0	0	1
123	2	0	1	0	2	0	0	0
124	2	0	1	0	2	0	0	1
125	2	0	1	0	2	1	0	0
126	2	0	1	0	2	1	0	1
127	2	0	1	0	3	0	0	0
128	2	0	1	0	3	0	0	1
129	2	0	1	0	3	1	0	0
130	2	0	1	0	3	1	0	1
131	2	0	1	1	1	0	0	0
132	2	0	1	1	1	0	0	1

#	n	ℓ	k	q	e	f	g	2S
133	2	0	1	1	1	0	1	0
134	2	0	1	1	1	0	1	1
135	2	0	1	1	2	0	0	0
136	2	0	1	1	2	0	0	1
137	2	0	1	1	2	0	1	0
138	2	0	1	1	2	0	1	1
139	2	0	1	1	2	1	0	0
140	2	0	1	1	2	1	0	1
141	2	0	1	1	2	1	1	0
142	2	0	1	1	2	1	1	1
143	2	0	1	1	3	0	0	0
144	2	0	1	1	3	0	0	1
145	2	0	1	1	3	0	1	0
146	2	0	1	1	3	0	1	1
147	2	0	1	1	3	1	0	0
148	2	0	1	1	3	1	0	1
149	2	0	1	1	3	1	1	0
150	2	0	1	1	3	1	1	1
151	2	0	2	0	1	0	0	0
152	2	0	2	0	1	0	0	1
153	2	0	2	0	1	0	0	2
154	2	0	2	0	2	0	0	0
155	2	0	2	0	2	0	0	1
156	2	0	2	0	2	0	0	2
157	2	0	2	0	2	1	0	0
158	2	0	2	0	2	1	0	1
159	2	0	2	0	2	1	0	2
160	2	0	2	0	3	0	0	0
161	2	0	2	0	3	0	0	1
162	2	0	2	0	3	0	0	2
163	2	0	2	0	3	1	0	0
164	2	0	2	0	3	1	0	1
165	2	0	2	0	3	1	0	2
166	2	0	2	1	1	0	0	0
167	2	0	2	1	1	0	0	1
168	2	0	2	1	1	0	0	2
169	2	0	2	1	1	0	1	0
170	2	0	2	1	1	0	1	1
171	2	0	2	1	1	0	1	2
172	2	0	2	1	2	0	0	0
173	2	0	2	1	2	0	0	1
174	2	0	2	1	2	0	0	2
175	2	0	2	1	2	0	1	0
176	2	0	2	1	2	0	1	1

#	n	ℓ	k	q	e	f	g	2S
177	2	0	2	1	2	0	1	2
178	2	0	2	1	2	1	0	0
179	2	0	2	1	2	1	0	1
180	2	0	2	1	2	1	0	2
181	2	0	2	1	2	1	1	0
182	2	0	2	1	2	1	1	1
183	2	0	2	1	2	1	1	2
184	2	0	2	1	3	0	0	0
185	2	0	2	1	3	0	0	1
186	2	0	2	1	3	0	0	2
187	2	0	2	1	3	0	1	0
188	2	0	2	1	3	0	1	1
189	2	0	2	1	3	0	1	2
190	2	0	2	1	3	1	0	0
191	2	0	2	1	3	1	0	1
192	2	0	2	1	3	1	0	2
193	2	0	2	1	3	1	1	0
194	2	0	2	1	3	1	1	1
195	2	0	2	1	3	1	1	2
196	2	0	2	2	1	0	0	0
197	2	0	2	2	1	0	0	1
198	2	0	2	2	1	0	0	2
199	2	0	2	2	1	0	1	0
200	2	0	2	2	1	0	1	1
201	2	0	2	2	1	0	1	2
202	2	0	2	2	1	0	2	0
203	2	0	2	2	1	0	2	1
204	2	0	2	2	1	0	2	2
205	2	0	2	2	2	0	0	0
206	2	0	2	2	2	0	0	1
207	2	0	2	2	2	0	0	2
208	2	0	2	2	2	0	1	0
209	2	0	2	2	2	0	1	1
210	2	0	2	2	2	0	1	2
211	2	0	2	2	2	0	2	0
212	2	0	2	2	2	0	2	1
213	2	0	2	2	2	0	2	2
214	2	0	2	2	2	1	0	0
215	2	0	2	2	2	1	0	1
216	2	0	2	2	2	1	0	2
217	2	0	2	2	2	1	1	0
218	2	0	2	2	2	1	1	1
219	2	0	2	2	2	1	1	2
220	2	0	2	2	2	1	2	0

#	n	ℓ	k	q	e	f	g	2S
221	2	0	2	2	2	1	2	1
222	2	0	2	2	2	1	2	2
223	2	0	2	2	3	0	0	0
224	2	0	2	2	3	0	0	1
225	2	0	2	2	3	0	0	2
226	2	0	2	2	3	0	1	0
227	2	0	2	2	3	0	1	1
228	2	0	2	2	3	0	1	2
229	2	0	2	2	3	0	2	0
230	2	0	2	2	3	0	2	1
231	2	0	2	2	3	0	2	2
232	2	0	2	2	3	1	0	0
233	2	0	2	2	3	1	0	1
234	2	0	2	2	3	1	0	2
235	2	0	2	2	3	1	1	0
236	2	0	2	2	3	1	1	1
237	2	0	2	2	3	1	1	2
238	2	0	2	2	3	1	2	0
239	2	0	2	2	3	1	2	1
240	2	0	2	2	3	1	2	2
241	2	1	1	0	1	0	0	0
242	2	1	1	0	1	0	0	1
243	2	1	1	0	2	0	0	0
244	2	1	1	0	2	0	0	1
245	2	1	1	0	2	1	0	0
246	2	1	1	0	2	1	0	1
247	2	1	1	0	3	0	0	0
248	2	1	1	0	3	0	0	1
249	2	1	1	0	3	1	0	0
250	2	1	1	0	3	1	0	1
251	2	1	1	1	1	0	0	0
252	2	1	1	1	1	0	0	1
253	2	1	1	1	1	0	1	0
254	2	1	1	1	1	0	1	1
255	2	1	1	1	2	0	0	0
256	2	1	1	1	2	0	0	1
257	2	1	1	1	2	0	1	0
258	2	1	1	1	2	0	1	1
259	2	1	1	1	2	1	0	0
260	2	1	1	1	2	1	0	1
261	2	1	1	1	2	1	1	0
262	2	1	1	1	2	1	1	1
263	2	1	1	1	3	0	0	0
264	2	1	1	1	3	0	0	1

#	n	ℓ	k	q	e	f	g	2S
265	2	1	1	1	3	0	1	0
266	2	1	1	1	3	0	1	1
267	2	1	1	1	3	1	0	0
268	2	1	1	1	3	1	0	1
269	2	1	1	1	3	1	1	0
270	2	1	1	1	3	1	1	1
271	2	1	2	0	1	0	0	0
272	2	1	2	0	1	0	0	1
273	2	1	2	0	1	0	0	2
274	2	1	2	0	2	0	0	0
275	2	1	2	0	2	0	0	1
276	2	1	2	0	2	0	0	2
277	2	1	2	0	2	1	0	0
278	2	1	2	0	2	1	0	1
279	2	1	2	0	2	1	0	2
280	2	1	2	0	3	0	0	0
281	2	1	2	0	3	0	0	1
282	2	1	2	0	3	0	0	2
283	2	1	2	0	3	1	0	0
284	2	1	2	0	3	1	0	1
285	2	1	2	0	3	1	0	2
286	2	1	2	1	1	0	0	0
287	2	1	2	1	1	0	0	1
288	2	1	2	1	1	0	0	2
289	2	1	2	1	1	0	1	0
290	2	1	2	1	1	0	1	1
291	2	1	2	1	1	0	1	2
292	2	1	2	1	2	0	0	0
293	2	1	2	1	2	0	0	1
294	2	1	2	1	2	0	0	2
295	2	1	2	1	2	0	1	0
296	2	1	2	1	2	0	1	1
297	2	1	2	1	2	0	1	2
298	2	1	2	1	2	1	0	0
299	2	1	2	1	2	1	0	1
300	2	1	2	1	2	1	0	2
301	2	1	2	1	2	1	1	0
302	2	1	2	1	2	1	1	1
303	2	1	2	1	2	1	1	2
304	2	1	2	1	3	0	0	0
305	2	1	2	1	3	0	0	1
306	2	1	2	1	3	0	0	2
307	2	1	2	1	3	0	1	0
308	2	1	2	1	3	0	1	1

#	n	ℓ	k	q	e	f	g	2S
309	2	1	2	1	3	0	1	2
310	2	1	2	1	3	1	0	0
311	2	1	2	1	3	1	0	1
312	2	1	2	1	3	1	0	2
313	2	1	2	1	3	1	1	0
314	2	1	2	1	3	1	1	1
315	2	1	2	1	3	1	1	2
316	2	1	2	2	1	0	0	0
317	2	1	2	2	1	0	0	1
318	2	1	2	2	1	0	0	2
319	2	1	2	2	1	0	1	0
320	2	1	2	2	1	0	1	1
321	2	1	2	2	1	0	1	2
322	2	1	2	2	1	0	2	0
323	2	1	2	2	1	0	2	1
324	2	1	2	2	1	0	2	2
325	2	1	2	2	2	0	0	0
326	2	1	2	2	2	0	0	1
327	2	1	2	2	2	0	0	2
328	2	1	2	2	2	0	1	0
329	2	1	2	2	2	0	1	1
330	2	1	2	2	2	0	1	2
331	2	1	2	2	2	0	2	0
332	2	1	2	2	2	0	2	1
333	2	1	2	2	2	0	2	2
334	2	1	2	2	2	1	0	0
335	2	1	2	2	2	1	0	1
336	2	1	2	2	2	1	0	2
337	2	1	2	2	2	1	1	0
338	2	1	2	2	2	1	1	1
339	2	1	2	2	2	1	1	2
340	2	1	2	2	2	1	2	0
341	2	1	2	2	2	1	2	1
342	2	1	2	2	2	1	2	2
343	2	1	2	2	3	0	0	0
344	2	1	2	2	3	0	0	1
345	2	1	2	2	3	0	0	2
346	2	1	2	2	3	0	1	0
347	2	1	2	2	3	0	1	1
348	2	1	2	2	3	0	1	2
349	2	1	2	2	3	0	2	0
350	2	1	2	2	3	0	2	1
351	2	1	2	2	3	0	2	2
352	2	1	2	2	3	1	0	0

#	n	ℓ	k	q	e	f	g	2S
353	2	1	2	2	3	1	0	1
354	2	1	2	2	3	1	0	2
355	2	1	2	2	3	1	1	0
356	2	1	2	2	3	1	1	1
357	2	1	2	2	3	1	1	2
358	2	1	2	2	3	1	2	0
359	2	1	2	2	3	1	2	1
360	2	1	2	2	3	1	2	2
361	2	1	3	0	1	0	0	0
362	2	1	3	0	1	0	0	1
363	2	1	3	0	1	0	0	2
364	2	1	3	0	1	0	0	3
365	2	1	3	0	2	0	0	0
366	2	1	3	0	2	0	0	1
367	2	1	3	0	2	0	0	2
368	2	1	3	0	2	0	0	3
369	2	1	3	0	2	1	0	0
370	2	1	3	0	2	1	0	1
371	2	1	3	0	2	1	0	2
372	2	1	3	0	2	1	0	3
373	2	1	3	0	3	0	0	0
374	2	1	3	0	3	0	0	1
375	2	1	3	0	3	0	0	2
376	2	1	3	0	3	0	0	3
377	2	1	3	0	3	1	0	0
378	2	1	3	0	3	1	0	1
379	2	1	3	0	3	1	0	2
380	2	1	3	0	3	1	0	3
381	2	1	3	1	1	0	0	0
382	2	1	3	1	1	0	0	1
383	2	1	3	1	1	0	0	2
384	2	1	3	1	1	0	0	3
385	2	1	3	1	1	0	1	0
386	2	1	3	1	1	0	1	1
387	2	1	3	1	1	0	1	2
388	2	1	3	1	1	0	1	3
389	2	1	3	1	2	0	0	0
390	2	1	3	1	2	0	0	1
391	2	1	3	1	2	0	0	2
392	2	1	3	1	2	0	0	3
393	2	1	3	1	2	0	1	0
394	2	1	3	1	2	0	1	1
395	2	1	3	1	2	0	1	2
396	2	1	3	1	2	0	1	3

#	n	ℓ	k	q	e	f	g	2S
397	2	1	3	1	2	1	0	0
398	2	1	3	1	2	1	0	1
399	2	1	3	1	2	1	0	2
400	2	1	3	1	2	1	0	3
401	2	1	3	1	2	1	1	0
402	2	1	3	1	2	1	1	1
403	2	1	3	1	2	1	1	2
404	2	1	3	1	2	1	1	3
405	2	1	3	1	3	0	0	0
406	2	1	3	1	3	0	0	1
407	2	1	3	1	3	0	0	2
408	2	1	3	1	3	0	0	3
409	2	1	3	1	3	0	1	0
410	2	1	3	1	3	0	1	1
411	2	1	3	1	3	0	1	2
412	2	1	3	1	3	0	1	3
413	2	1	3	1	3	1	0	0
414	2	1	3	1	3	1	0	1
415	2	1	3	1	3	1	0	2
416	2	1	3	1	3	1	0	3
417	2	1	3	1	3	1	1	0
418	2	1	3	1	3	1	1	1
419	2	1	3	1	3	1	1	2
420	2	1	3	1	3	1	1	3
421	2	1	3	2	1	0	0	0
422	2	1	3	2	1	0	0	1
423	2	1	3	2	1	0	0	2
424	2	1	3	2	1	0	0	3
425	2	1	3	2	1	0	1	0
426	2	1	3	2	1	0	1	1
427	2	1	3	2	1	0	1	2
428	2	1	3	2	1	0	1	3
429	2	1	3	2	1	0	2	0
430	2	1	3	2	1	0	2	1
431	2	1	3	2	1	0	2	2
432	2	1	3	2	1	0	2	3
433	2	1	3	2	2	0	0	0
434	2	1	3	2	2	0	0	1
435	2	1	3	2	2	0	0	2
436	2	1	3	2	2	0	0	3
437	2	1	3	2	2	0	1	0
438	2	1	3	2	2	0	1	1
439	2	1	3	2	2	0	1	2
440	2	1	3	2	2	0	1	3

#	n	ℓ	k	q	e	f	g	2S
441	2	1	3	2	2	0	2	0
442	2	1	3	2	2	0	2	1
443	2	1	3	2	2	0	2	2
444	2	1	3	2	2	0	2	3
445	2	1	3	2	2	1	0	0
446	2	1	3	2	2	1	0	1
447	2	1	3	2	2	1	0	2
448	2	1	3	2	2	1	0	3
449	2	1	3	2	2	1	1	0
450	2	1	3	2	2	1	1	1
451	2	1	3	2	2	1	1	2
452	2	1	3	2	2	1	1	3
453	2	1	3	2	2	1	2	0
454	2	1	3	2	2	1	2	1
455	2	1	3	2	2	1	2	2
456	2	1	3	2	2	1	2	3
457	2	1	3	2	3	0	0	0
458	2	1	3	2	3	0	0	1
459	2	1	3	2	3	0	0	2
460	2	1	3	2	3	0	0	3
461	2	1	3	2	3	0	1	0
462	2	1	3	2	3	0	1	1
463	2	1	3	2	3	0	1	2
464	2	1	3	2	3	0	1	3
465	2	1	3	2	3	0	2	0
466	2	1	3	2	3	0	2	1
467	2	1	3	2	3	0	2	2
468	2	1	3	2	3	0	2	3
469	2	1	3	2	3	1	0	0
470	2	1	3	2	3	1	0	1
471	2	1	3	2	3	1	0	2
472	2	1	3	2	3	1	0	3
473	2	1	3	2	3	1	1	0
474	2	1	3	2	3	1	1	1
475	2	1	3	2	3	1	1	2
476	2	1	3	2	3	1	1	3
477	2	1	3	2	3	1	2	0
478	2	1	3	2	3	1	2	1
479	2	1	3	2	3	1	2	2
480	2	1	3	2	3	1	2	3
481	2	1	3	3	1	0	0	0
482	2	1	3	3	1	0	0	1
483	2	1	3	3	1	0	0	2
484	2	1	3	3	1	0	0	3

#	n	ℓ	k	q	e	f	g	2S
485	2	1	3	3	1	0	1	0
486	2	1	3	3	1	0	1	1
487	2	1	3	3	1	0	1	2
488	2	1	3	3	1	0	1	3
489	2	1	3	3	1	0	2	0
490	2	1	3	3	1	0	2	1
491	2	1	3	3	1	0	2	2
492	2	1	3	3	1	0	2	3
493	2	1	3	3	2	0	0	0
494	2	1	3	3	2	0	0	1
495	2	1	3	3	2	0	0	2
496	2	1	3	3	2	0	0	3
497	2	1	3	3	2	0	1	0
498	2	1	3	3	2	0	1	1
499	2	1	3	3	2	0	1	2
500	2	1	3	3	2	0	1	3
501	2	1	3	3	2	0	2	0
502	2	1	3	3	2	0	2	1
503	2	1	3	3	2	0	2	2
504	2	1	3	3	2	0	2	3
505	2	1	3	3	2	1	0	0
506	2	1	3	3	2	1	0	1
507	2	1	3	3	2	1	0	2
508	2	1	3	3	2	1	0	3
509	2	1	3	3	2	1	1	0
510	2	1	3	3	2	1	1	1
511	2	1	3	3	2	1	1	2
512	2	1	3	3	2	1	1	3
513	2	1	3	3	2	1	2	0
514	2	1	3	3	2	1	2	1
515	2	1	3	3	2	1	2	2
516	2	1	3	3	2	1	2	3
517	2	1	3	3	2	1	3	0
518	2	1	3	3	2	1	3	1
519	2	1	3	3	2	1	3	2
520	2	1	3	3	2	1	3	3
521	2	1	3	3	3	0	0	0
522	2	1	3	3	3	0	0	1
523	2	1	3	3	3	0	0	2
524	2	1	3	3	3	0	0	3
525	2	1	3	3	3	0	1	0
526	2	1	3	3	3	0	1	1
527	2	1	3	3	3	0	1	2
528	2	1	3	3	3	0	1	3

#	n	ℓ	k	q	e	f	g	2S
529	2	1	3	3	3	0	2	0
530	2	1	3	3	3	0	2	1
531	2	1	3	3	3	0	2	2
532	2	1	3	3	3	0	2	3
533	2	1	3	3	3	1	0	0
534	2	1	3	3	3	1	0	1
535	2	1	3	3	3	1	0	2
536	2	1	3	3	3	1	0	3
537	2	1	3	3	3	1	1	0
538	2	1	3	3	3	1	1	1
539	2	1	3	3	3	1	1	2
540	2	1	3	3	3	1	1	3
541	2	1	3	3	3	1	2	0
542	2	1	3	3	3	1	2	1
543	2	1	3	3	3	1	2	2
544	2	1	3	3	3	1	2	3
545	2	1	3	3	3	1	3	0
546	2	1	3	3	3	1	3	1
547	2	1	3	3	3	1	3	2
548	2	1	3	3	3	1	3	3
549	3	0	1	0	1	0	0	0
550	3	0	1	0	1	0	0	1
551	3	0	1	0	2	0	0	0
552	3	0	1	0	2	0	0	1
553	3	0	1	0	2	1	0	0
554	3	0	1	0	2	1	0	1
555	3	0	1	0	3	0	0	0
556	3	0	1	0	3	0	0	1
557	3	0	1	0	3	1	0	0
558	3	0	1	0	3	1	0	1
559	3	0	1	1	1	0	0	0
560	3	0	1	1	1	0	0	1
561	3	0	1	1	1	0	1	0
562	3	0	1	1	1	0	1	1
563	3	0	1	1	2	0	0	0
564	3	0	1	1	2	0	0	1
565	3	0	1	1	2	0	1	0
566	3	0	1	1	2	0	1	1
567	3	0	1	1	2	1	0	0
568	3	0	1	1	2	1	0	1
569	3	0	1	1	2	1	1	0
570	3	0	1	1	2	1	1	1
571	3	0	1	1	3	0	0	0
572	3	0	1	1	3	0	0	1

#	n	ℓ	k	q	e	f	g	2S
573	3	0	1	1	3	0	1	0
574	3	0	1	1	3	0	1	1
575	3	0	1	1	3	1	0	0
576	3	0	1	1	3	1	0	1
577	3	0	1	1	3	1	1	0
578	3	0	1	1	3	1	1	1
579	3	0	2	0	1	0	0	0
580	3	0	2	0	1	0	0	1
581	3	0	2	0	1	0	0	2
582	3	0	2	0	2	0	0	0
583	3	0	2	0	2	0	0	1
584	3	0	2	0	2	0	0	2
585	3	0	2	0	2	1	0	0
586	3	0	2	0	2	1	0	1
587	3	0	2	0	2	1	0	2
588	3	0	2	0	3	0	0	0
589	3	0	2	0	3	0	0	1
590	3	0	2	0	3	0	0	2
591	3	0	2	0	3	1	0	0
592	3	0	2	0	3	1	0	1
593	3	0	2	0	3	1	0	2
594	3	0	2	1	1	0	0	0
595	3	0	2	1	1	0	0	1
596	3	0	2	1	1	0	0	2
597	3	0	2	1	1	0	1	0
598	3	0	2	1	1	0	1	1
599	3	0	2	1	1	0	1	2
600	3	0	2	1	2	0	0	0
601	3	0	2	1	2	0	0	1
602	3	0	2	1	2	0	0	2
603	3	0	2	1	2	0	1	0
604	3	0	2	1	2	0	1	1
605	3	0	2	1	2	0	1	2
606	3	0	2	1	2	1	0	0
607	3	0	2	1	2	1	0	1
608	3	0	2	1	2	1	0	2
609	3	0	2	1	2	1	1	0
610	3	0	2	1	2	1	1	1
611	3	0	2	1	2	1	1	2
612	3	0	2	1	3	0	0	0
613	3	0	2	1	3	0	0	1
614	3	0	2	1	3	0	0	2
615	3	0	2	1	3	0	1	0
616	3	0	2	1	3	0	1	1

#	n	ℓ	k	q	e	f	g	2S
617	3	0	2	1	3	0	1	2
618	3	0	2	1	3	1	0	0
619	3	0	2	1	3	1	0	1
620	3	0	2	1	3	1	0	2
621	3	0	2	1	3	1	1	0
622	3	0	2	1	3	1	1	1
623	3	0	2	1	3	1	1	2
624	3	0	2	2	1	0	0	0
625	3	0	2	2	1	0	0	1
626	3	0	2	2	1	0	0	2
627	3	0	2	2	1	0	1	0
628	3	0	2	2	1	0	1	1
629	3	0	2	2	1	0	1	2
630	3	0	2	2	1	0	2	0
631	3	0	2	2	1	0	2	1
632	3	0	2	2	1	0	2	2
633	3	0	2	2	2	0	0	0
634	3	0	2	2	2	0	0	1
635	3	0	2	2	2	0	0	2
636	3	0	2	2	2	0	1	0
637	3	0	2	2	2	0	1	1
638	3	0	2	2	2	0	1	2
639	3	0	2	2	2	0	2	0
640	3	0	2	2	2	0	2	1
641	3	0	2	2	2	0	2	2
642	3	0	2	2	2	1	0	0
643	3	0	2	2	2	1	0	1
644	3	0	2	2	2	1	0	2
645	3	0	2	2	2	1	1	0
646	3	0	2	2	2	1	1	1
647	3	0	2	2	2	1	1	2
648	3	0	2	2	2	1	2	0
649	3	0	2	2	2	1	2	1
650	3	0	2	2	2	1	2	2
651	3	0	2	2	3	0	0	0
652	3	0	2	2	3	0	0	1
653	3	0	2	2	3	0	0	2
654	3	0	2	2	3	0	1	0
655	3	0	2	2	3	0	1	1
656	3	0	2	2	3	0	1	2
657	3	0	2	2	3	0	2	0
658	3	0	2	2	3	0	2	1
659	3	0	2	2	3	0	2	2
660	3	0	2	2	3	1	0	0

#	n	ℓ	k	q	e	f	g	2S
661	3	0	2	2	3	1	0	1
662	3	0	2	2	3	1	0	2
663	3	0	2	2	3	1	1	0
664	3	0	2	2	3	1	1	1
665	3	0	2	2	3	1	1	2
666	3	0	2	2	3	1	2	0
667	3	0	2	2	3	1	2	1
668	3	0	2	2	3	1	2	2
669	3	1	1	0	1	0	0	0
670	3	1	1	0	1	0	0	1
671	3	1	1	0	2	0	0	0
672	3	1	1	0	2	0	0	1
673	3	1	1	0	2	1	0	0
674	3	1	1	0	2	1	0	1
675	3	1	1	0	3	0	0	0
676	3	1	1	0	3	0	0	1
677	3	1	1	0	3	1	0	0
678	3	1	1	0	3	1	0	1
679	3	1	1	1	1	0	0	0
680	3	1	1	1	1	0	0	1
681	3	1	1	1	1	0	1	0
682	3	1	1	1	1	0	1	1
683	3	1	1	1	2	0	0	0
684	3	1	1	1	2	0	0	1
685	3	1	1	1	2	0	1	0
686	3	1	1	1	2	0	1	1
687	3	1	1	1	2	1	0	0
688	3	1	1	1	2	1	0	1
689	3	1	1	1	2	1	1	0
690	3	1	1	1	2	1	1	1
691	3	1	1	1	3	0	0	0
692	3	1	1	1	3	0	0	1
693	3	1	1	1	3	0	1	0
694	3	1	1	1	3	0	1	1
695	3	1	1	1	3	1	0	0
696	3	1	1	1	3	1	0	1
697	3	1	1	1	3	1	1	0
698	3	1	1	1	3	1	1	1
699	3	1	2	0	1	0	0	0
700	3	1	2	0	1	0	0	1
701	3	1	2	0	1	0	0	2
702	3	1	2	0	2	0	0	0
703	3	1	2	0	2	0	0	1
704	3	1	2	0	2	0	0	2

#	n	ℓ	k	q	e	f	g	2S
705	3	1	2	0	2	1	0	0
706	3	1	2	0	2	1	0	1
707	3	1	2	0	2	1	0	2
708	3	1	2	0	3	0	0	0
709	3	1	2	0	3	0	0	1
710	3	1	2	0	3	0	0	2
711	3	1	2	0	3	1	0	0
712	3	1	2	0	3	1	0	1
713	3	1	2	0	3	1	0	2
714	3	1	2	1	1	0	0	0
715	3	1	2	1	1	0	0	1
716	3	1	2	1	1	0	0	2
717	3	1	2	1	1	0	1	0
718	3	1	2	1	1	0	1	1
719	3	1	2	1	1	0	1	2
720	3	1	2	1	2	0	0	0
721	3	1	2	1	2	0	0	1
722	3	1	2	1	2	0	0	2
723	3	1	2	1	2	0	1	0
724	3	1	2	1	2	0	1	1
725	3	1	2	1	2	0	1	2
726	3	1	2	1	2	1	0	0
727	3	1	2	1	2	1	0	1
728	3	1	2	1	2	1	0	2
729	3	1	2	1	2	1	1	0
730	3	1	2	1	2	1	1	1
731	3	1	2	1	2	1	1	2
732	3	1	2	1	3	0	0	0
733	3	1	2	1	3	0	0	1
734	3	1	2	1	3	0	0	2
735	3	1	2	1	3	0	1	0
736	3	1	2	1	3	0	1	1
737	3	1	2	1	3	0	1	2
738	3	1	2	1	3	1	0	0
739	3	1	2	1	3	1	0	1
740	3	1	2	1	3	1	0	2
741	3	1	2	1	3	1	1	0
742	3	1	2	1	3	1	1	1
743	3	1	2	1	3	1	1	2
744	3	1	2	2	1	0	0	0
745	3	1	2	2	1	0	0	1
746	3	1	2	2	1	0	0	2
747	3	1	2	2	1	0	1	0
748	3	1	2	2	1	0	1	1

#	n	ℓ	k	q	e	f	g	2S
749	3	1	2	2	1	0	1	2
750	3	1	2	2	1	0	2	0
751	3	1	2	2	1	0	2	1
752	3	1	2	2	1	0	2	2
753	3	1	2	2	2	0	0	0
754	3	1	2	2	2	0	0	1
755	3	1	2	2	2	0	0	2
756	3	1	2	2	2	0	1	0
757	3	1	2	2	2	0	1	1
758	3	1	2	2	2	0	1	2
759	3	1	2	2	2	0	2	0
760	3	1	2	2	2	0	2	1
761	3	1	2	2	2	0	2	2
762	3	1	2	2	2	1	0	0
763	3	1	2	2	2	1	0	1
764	3	1	2	2	2	1	0	2
765	3	1	2	2	2	1	1	0
766	3	1	2	2	2	1	1	1
767	3	1	2	2	2	1	1	2
768	3	1	2	2	2	1	2	0
769	3	1	2	2	2	1	2	1
770	3	1	2	2	2	1	2	2
771	3	1	2	2	3	0	0	0
772	3	1	2	2	3	0	0	1
773	3	1	2	2	3	0	0	2
774	3	1	2	2	3	0	1	0
775	3	1	2	2	3	0	1	1
776	3	1	2	2	3	0	1	2
777	3	1	2	2	3	0	2	0
778	3	1	2	2	3	0	2	1
779	3	1	2	2	3	0	2	2
780	3	1	2	2	3	1	0	0
781	3	1	2	2	3	1	0	1
782	3	1	2	2	3	1	0	2
783	3	1	2	2	3	1	1	0
784	3	1	2	2	3	1	1	1
785	3	1	2	2	3	1	1	2
786	3	1	2	2	3	1	2	0
787	3	1	2	2	3	1	2	1
788	3	1	2	2	3	1	2	2
789	3	1	3	0	1	0	0	0
790	3	1	3	0	1	0	0	1
791	3	1	3	0	1	0	0	2
792	3	1	3	0	1	0	0	3

#	n	ℓ	k	q	e	f	g	2S
793	3	1	3	0	2	0	0	0
794	3	1	3	0	2	0	0	1
795	3	1	3	0	2	0	0	2
796	3	1	3	0	2	0	0	3
797	3	1	3	0	2	1	0	0
798	3	1	3	0	2	1	0	1
799	3	1	3	0	2	1	0	2
800	3	1	3	0	2	1	0	3
801	3	1	3	0	3	0	0	0
802	3	1	3	0	3	0	0	1
803	3	1	3	0	3	0	0	2
804	3	1	3	0	3	0	0	3
805	3	1	3	0	3	1	0	0
806	3	1	3	0	3	1	0	1
807	3	1	3	0	3	1	0	2
808	3	1	3	0	3	1	0	3
809	3	1	3	1	1	0	0	0
810	3	1	3	1	1	0	0	1
811	3	1	3	1	1	0	0	2
812	3	1	3	1	1	0	0	3
813	3	1	3	1	1	0	1	0
814	3	1	3	1	1	0	1	1
815	3	1	3	1	1	0	1	2
816	3	1	3	1	1	0	1	3
817	3	1	3	1	2	0	0	0
818	3	1	3	1	2	0	0	1
819	3	1	3	1	2	0	0	2
820	3	1	3	1	2	0	0	3
821	3	1	3	1	2	0	1	0
822	3	1	3	1	2	0	1	1
823	3	1	3	1	2	0	1	2
824	3	1	3	1	2	0	1	3
825	3	1	3	1	2	1	0	0
826	3	1	3	1	2	1	0	1
827	3	1	3	1	2	1	0	2
828	3	1	3	1	2	1	0	3
829	3	1	3	1	2	1	1	0
830	3	1	3	1	2	1	1	1
831	3	1	3	1	2	1	1	2
832	3	1	3	1	2	1	1	3
833	3	1	3	1	3	0	0	0
834	3	1	3	1	3	0	0	1
835	3	1	3	1	3	0	0	2
836	3	1	3	1	3	0	0	3

#	n	ℓ	k	q	e	f	g	2S
837	3	1	3	1	3	0	1	0
838	3	1	3	1	3	0	1	1
839	3	1	3	1	3	0	1	2
840	3	1	3	1	3	0	1	3
841	3	1	3	1	3	1	0	0
842	3	1	3	1	3	1	0	1
843	3	1	3	1	3	1	0	2
844	3	1	3	1	3	1	0	3
845	3	1	3	1	3	1	1	0
846	3	1	3	1	3	1	1	1
847	3	1	3	1	3	1	1	2
848	3	1	3	1	3	1	1	3
849	3	1	3	2	1	0	0	0
850	3	1	3	2	1	0	0	1
851	3	1	3	2	1	0	0	2
852	3	1	3	2	1	0	0	3
853	3	1	3	2	1	0	1	0
854	3	1	3	2	1	0	1	1
855	3	1	3	2	1	0	1	2
856	3	1	3	2	1	0	1	3
857	3	1	3	2	1	0	2	0
858	3	1	3	2	1	0	2	1
859	3	1	3	2	1	0	2	2
860	3	1	3	2	1	0	2	3
861	3	1	3	2	2	0	0	0
862	3	1	3	2	2	0	0	1
863	3	1	3	2	2	0	0	2
864	3	1	3	2	2	0	0	3
865	3	1	3	2	2	0	1	0
866	3	1	3	2	2	0	1	1
867	3	1	3	2	2	0	1	2
868	3	1	3	2	2	0	1	3
869	3	1	3	2	2	0	2	0
870	3	1	3	2	2	0	2	1
871	3	1	3	2	2	0	2	2
872	3	1	3	2	2	0	2	3
873	3	1	3	2	2	1	0	0
874	3	1	3	2	2	1	0	1
875	3	1	3	2	2	1	0	2
876	3	1	3	2	2	1	0	3
877	3	1	3	2	2	1	1	0
878	3	1	3	2	2	1	1	1
879	3	1	3	2	2	1	1	2
880	3	1	3	2	2	1	1	3

#	n	ℓ	k	q	e	f	g	2S
881	3	1	3	2	2	1	2	0
882	3	1	3	2	2	1	2	1
883	3	1	3	2	2	1	2	2
884	3	1	3	2	2	1	2	3
885	3	1	3	2	3	0	0	0
886	3	1	3	2	3	0	0	1
887	3	1	3	2	3	0	0	2
888	3	1	3	2	3	0	0	3
889	3	1	3	2	3	0	1	0
890	3	1	3	2	3	0	1	1
891	3	1	3	2	3	0	1	2
892	3	1	3	2	3	0	1	3
893	3	1	3	2	3	0	2	0
894	3	1	3	2	3	0	2	1
895	3	1	3	2	3	0	2	2
896	3	1	3	2	3	0	2	3
897	3	1	3	2	3	1	0	0
898	3	1	3	2	3	1	0	1
899	3	1	3	2	3	1	0	2
900	3	1	3	2	3	1	0	3
901	3	1	3	2	3	1	1	0
902	3	1	3	2	3	1	1	1
903	3	1	3	2	3	1	1	2
904	3	1	3	2	3	1	1	3
905	3	1	3	2	3	1	2	0
906	3	1	3	2	3	1	2	1
907	3	1	3	2	3	1	2	2
908	3	1	3	2	3	1	2	3
909	3	1	3	3	1	0	0	0
910	3	1	3	3	1	0	0	1
911	3	1	3	3	1	0	0	2
912	3	1	3	3	1	0	0	3
913	3	1	3	3	1	0	1	0
914	3	1	3	3	1	0	1	1
915	3	1	3	3	1	0	1	2
916	3	1	3	3	1	0	1	3
917	3	1	3	3	1	0	2	0
918	3	1	3	3	1	0	2	1
919	3	1	3	3	1	0	2	2
920	3	1	3	3	1	0	2	3
921	3	1	3	3	2	0	0	0
922	3	1	3	3	2	0	0	1
923	3	1	3	3	2	0	0	2
924	3	1	3	3	2	0	0	3

#	n	ℓ	k	q	e	f	g	2S
925	3	1	3	3	2	0	1	0
926	3	1	3	3	2	0	1	1
927	3	1	3	3	2	0	1	2
928	3	1	3	3	2	0	1	3
929	3	1	3	3	2	0	2	0
930	3	1	3	3	2	0	2	1
931	3	1	3	3	2	0	2	2
932	3	1	3	3	2	0	2	3
933	3	1	3	3	2	1	0	0
934	3	1	3	3	2	1	0	1
935	3	1	3	3	2	1	0	2
936	3	1	3	3	2	1	0	3
937	3	1	3	3	2	1	1	0
938	3	1	3	3	2	1	1	1
939	3	1	3	3	2	1	1	2
940	3	1	3	3	2	1	1	3
941	3	1	3	3	2	1	2	0
942	3	1	3	3	2	1	2	1
943	3	1	3	3	2	1	2	2
944	3	1	3	3	2	1	2	3
945	3	1	3	3	2	1	3	0
946	3	1	3	3	2	1	3	1
947	3	1	3	3	2	1	3	2
948	3	1	3	3	2	1	3	3
949	3	1	3	3	3	0	0	0
950	3	1	3	3	3	0	0	1
951	3	1	3	3	3	0	0	2
952	3	1	3	3	3	0	0	3
953	3	1	3	3	3	0	1	0
954	3	1	3	3	3	0	1	1
955	3	1	3	3	3	0	1	2
956	3	1	3	3	3	0	1	3
957	3	1	3	3	3	0	2	0
958	3	1	3	3	3	0	2	1
959	3	1	3	3	3	0	2	2
960	3	1	3	3	3	0	2	3
961	3	1	3	3	3	1	0	0
962	3	1	3	3	3	1	0	1
963	3	1	3	3	3	1	0	2
964	3	1	3	3	3	1	0	3
965	3	1	3	3	3	1	1	0
966	3	1	3	3	3	1	1	1
967	3	1	3	3	3	1	1	2
968	3	1	3	3	3	1	1	3

#	n	ℓ	k	q	e	f	g	2S
969	3	1	3	3	3	1	2	0
970	3	1	3	3	3	1	2	1
971	3	1	3	3	3	1	2	2
972	3	1	3	3	3	1	2	3
973	3	1	3	3	3	1	3	0
974	3	1	3	3	3	1	3	1
975	3	1	3	3	3	1	3	2
976	3	1	3	3	3	1	3	3

976 rows. F(1) = 976, F(−1) = 2, E = 0, density 0.1412 of a box of 6,912.

The periodic table — all 90 cells, (period, group)

period	groups occupied
1	1, 18
2	1, 2, 13, 14, 15, 16, 17, 18
3	1, 2, 13, 14, 15, 16, 17, 18
4	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18
5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18
6	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18
7	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18

90 cells in a box of 126. E = 36 — and the 36 are the cells the coordinates admit and no element occupies, which must be supplied from outside.

Janet — all 118, (n+ℓ, Z)

n+ℓ	atomic numbers
1	1–2 (2)
2	3–4 (2)
3	5–12 (8)
4	13–20 (8)
5	21–38 (18)
6	39–56 (18)
7	57–88 (32)
8	89–118 (30)

118 cells, E = 0. The same 118 elements as the table above, reordered by n+ℓ.

The calendar — 365 cells, (month, day)

month	days	month	days
January	31	July	31
February	28	August	31
March	31	September	30
April	30	October	31
May	31	November	30
June	30	December	31

365 cells in a box of 372. $E = 7$ — February’s missing 29th, 30th and 31st, and the four thirty-day months’ 31sts. **$365 = 5 \cdot 73$, and 73 exceeds every rung**, so it is genuinely constrained.

A box ordering — all 35, $l \geq w \geq h$

$l \geq w \geq h$
0·0·0 1·0·0 1·1·0 1·1·1 2·0·0 2·1·0 2·1·1
2·2·0 2·2·1 2·2·2 3·0·0 3·1·0 3·1·1 3·2·0
3·2·1 3·2·2 3·3·0 3·3·1 3·3·2 3·3·3 4·0·0
4·1·0 4·1·1 4·2·0 4·2·1 4·2·2 4·3·0 4·3·1
4·3·2 4·3·3 4·4·0 4·4·1 4·4·2 4·4·3 4·4·4

35 cells in a box of 125, $E = 0$. $35 = 5 \cdot 7$ and 7 exceeds the rung of 5: a real constraint.

VIII · WHAT EACH INDEX IS, AND WHAT ONLY IT CONTRIBUTES

Λ is the only index here built from physics. The rest are built from Λ , from the act of building Λ , or from drawing the same subject another way. *This section states for each one: what it is, where it comes from in the published record, and the one thing it contributes to the atomic index that no other index can.*

Λ — the atomic index

What it is. Eight integer coordinates ($n, \ell, k, q, e, f, g, 2S$) under eight inequalities, 976 cells, $E = 0$.

Where it comes from. The constraints are not this work's: $\ell \leq n-1$ is the hydrogen solution (Bohr 1913; Schrödinger 1926); $k \leq 4\ell+2$ is Stoner's subshell capacity (1924) made exclusive by Pauli (1925); $2S \leq k$ is Pauli with Hund's first rule (1925). **Λ assembles them; it introduces none.**

What only it contributes. *The transfer.* Λ is the only index here whose cells are MOVES rather than positions — a cell carries a source subshell, a target subshell and a count transferred. **Everything about composition, time columns and categories follows from that and from nothing else in this compendium.**

The tower $\Lambda_8 \dots \Lambda_{13}$

What it is. Λ with the coupling quantum numbers adjoined one at a time — seniority, the core J, K, then J. Cell counts 976, 1654, 2535, 13585, 70905, 199130, $E = 0$ at every stage.

Where it comes from. Each axis is Racah or Condon & Shortley: seniority is Racah, *Theory of complex spectra III*, Phys. Rev. **63** (1943) 367–382; the core J and the recoupling coefficients are Racah, Phys. Rev. **62** (1942) 438–462; the jK and LK schemes are Condon & Shortley (1935), ch. X.

What only it contributes. *The price of an axis.* The tower is the only construction here that adds one coordinate at a time and measures what each costs and buys — **cycle rank, composability, exactness, and the point at which the tree becomes a graph.** No single index can show that; it takes a sequence.

The periodic table — (period, group)

What it is. 90 cells in a box of 126, $E = 36$.

Where it comes from. Mendeleev, *Über die Beziehungen der Eigenschaften zu den Atomgewichten der Elemente*, Z. Chem. **12** (1869) 405–406. At least a thousand periodic systems have been published since (van Spronsen; Scerri, *A Tale of Seven Scientists*, 2016).

What only it contributes. *The control.* It is the one index whose coordinates were fixed by other people for other reasons, long before this work, and which nonetheless supplies **118 elements and 18,288 constraint tests with zero failures. An index built by the author cannot be a control; this one can.**

Janet's left-step table — ($n+\ell, Z$)

What it is. The same 118 elements, ordered on $n+\ell$. 118 cells in a box of 944, $E = 0$.

Where it comes from. Charles Janet, *Considérations sur la structure du noyau de l'atome*, Beauvais (1929), with the table first published in **1928**. Janet recognised the $(n+\ell)$ rule before Madelung, who arrived at it around 1926 and did not publish until 1936. The shell-length sequence 2, 2, 8, 8, 18, 18, 32, 32 follows from the rule by the Klechkovski–Hakala formulas.

What only it contributes. That E is coordinate-relative, demonstrated on one subject rather than argued. The periodic table and Janet index THE SAME 118 elements and give $E = 36$ and $E = 0$. **No pair of indexes anywhere else in this work makes that point as cleanly, because no other pair shares its subject exactly.**

And an open problem sits underneath it. Why the $n+\ell$ ordering holds is unresolved — Löwdin's challenge, still open; see Allen & Knight, *The Löwdin challenge: origin of the $n+l$, n (Madelung) rule*, Int. J. Quantum Chem. **90** (2003) 80–88. **This work uses the ordering and does not explain it**, and the orbital-collapse result (registers 1187–1190) is a measurement against Janet's boundaries, not a derivation of them.

The nuclide chart — (Z, N)

What it is. The measured nuclides indexed by proton and neutron count, $E = 9$.

Where it comes from. The chart is Segre's, in use since the 1940s; the values are the AME2020 evaluation.

What only it contributes. A defect whose cells are identifiable physics. The nine cells are the mass formula's pairing and clustering terms. **It is the only index here where $E > 0$ and every missing cell can be named**, which is what makes the defect a measurement rather than a score.

The calendar — (month, day)

What it is. 365 cells in a box of 372, $E = 7$.

Where it comes from. The Gregorian reform of 1582, on the Julian arrangement of 46 BC. *The month lengths are a political inheritance, not a natural one.*

What only it contributes. A subject with no physics in it at all. The calendar shows the operator working on an object whose constraints are entirely conventional, **and $365 = 5 \cdot 73$ with $E = 7$ is the price of keeping January first**. No other index here separates the method from the physics so completely.

The box ordering and the chessboard

What they are. $\ell \geq w \geq h$ over five values: 35 cells in 125, $E = 0$. And (rank, file): 64 cells in 64, $E = 0$.

What only they contribute. *The floor and the ceiling.* The chessboard is a full product — $E = 0$ because there is no constraint at all. The box ordering is a genuine constraint that still closes. **Between them they show that $E = 0$ carries information only when the ambient box exceeds the cells**, which is why the compendium reports box alongside every defect.

The electromagnetic quotient

What it is. Λ_g under the dipole selection rules: $|\Delta\ell| = 1$ and $\Delta S = 0$.

Where it comes from. The parity rule is Laporte, Z. Phys. **23** (1924) 135; the spin rule is Russell & Saunders, Astrophys. J. **61** (1925) 38; both have their group-theoretic ground in Wigner, Z. Phys. **43** (1927) 624.

What only it contributes. *That a selection rule is a QUOTIENT and not an extension.* Adjoining the multipole and ΔS as coordinates gives $E = 3$; taking the image gives $E = 0$. **It is the only index here that demonstrates the difference between adding a coordinate and dividing by one.**

Λ_{spectra} — the channel index

What it is. (Z , charge, ℓ , $2S+1$) over 104,832 cells, with a defect for each.

Where it comes from. Quantum defect theory is Seaton, MNRAS **118** (1958) 504–518, and Rep. Prog. Phys. **46** (1983) 167–257. *A published survey of the same object exists:* Theodosiou, Inokuti & Manson, At. Data Nucl. Data Tables **35** (1986) 473–486, Hartree–Slater, for all ionisation stages of all ions with $Z \leq 50$.

What only it contributes. *Values.* Every other index here holds cells and asks whether they close. Λ_{spectra} holds cells that carry **NUMBERS**, which is why it is the only index on which the Method equation’s metric half — E_W — has anything to measure at all (registers 1196–1199).

The violation index

What it is. The companion paper’s object over nine and fifteen letters, with a core of three conditions whose joint failure is irreducible.

Where it comes from. The conditions are physics — the null energy condition (Penrose 1965), ghost states (Pais & Uhlenbeck 1950), the equations of motion. *The core is a MINIMAL UNSATISFIABLE SUBSET* in the sense of Chinneck & Dravnieks, *Locating minimal infeasible constraint sets in linear programs*, ORSA J. Comput. **3** (1991) 157–168.

What only it contributes. *An index this work reasons about without holding.* Its cells are not printed anywhere. **It is the only test of whether the method says anything when the object is out of reach**, and the answer is that it locates a three-condition core from published summary numbers alone.

Λ_{phys} — the parameter index

What it is. Every number the work takes from physics, on four coordinates — kind (exact $\rightarrow$ fitted), source (mathematics $\rightarrow$ this work), domain (universal $\rightarrow$ one species, and the three-body problem’s own range), and how many registered objects rest on it. **27 parameters** (22 before Chapters 35 and 36 added c , the three-body masses, E , L and G).

Where it comes from. The parameters are Rydberg 1890, Bohr 1913, Stoner 1924, Pauli 1925, Hund 1925, Fermi 1928, Hartree 1928, Janet 1928, Mayer & Mayer 1933, Goepfert-Mayer 1941, Seaton 1958, Edlén 1964, Griffin, Andrew & Cowan 1969, CODATA 2018 — and eight numbers fitted here. *The Physics Compendium holds them in full, each with a definition, a dated source and a failure mode.*

What only it contributes. *Where the work would break first.* Every other index here asks whether a set of cells closes. Λ_{phys} asks **what the whole construction rests on, and the answer is a shape**: eight of the twenty-seven parameters are this work’s own, **none of them is universal**, seven of the eight hold only in a region or for one species — and, on the one rule the Physics Compendium now prints, the two carrying the most objects are the lattice’s own exact bounds, the angular-momentum bound (72 objects) and the subshell capacity (66), with the aufbau ordering (18) next; the earlier reading — the ionisation limit 25, the aufbau ordering 19, the Janet boundary 15, on a rule never printed — stands in that compendium as the prior state (register 1740).

And three of its failure modes are MEASURED rather than anticipated — Seaton's ratio at 1.25 where the dipole term gives 1.00, the Thomas–Fermi exponent rising from 0.84 to 1.52 with charge against a single fitted 0.494, and the exchange coefficient whose fitted sign is opposite to the measured one.

The languages

What they are. Six readings of one index — order, analysis, algebra, geometry, information, statistics — with documentary as a seventh that has no operator.

Where they come from. Each operator is standard in its own field: closure from Moore (1910), max-entropy on marginals from Deming & Stephan (1940) and Csiszár (1975), conditional independence from Dawid (1979).

What only they contribute. *That translation is re-coordinatisation and E is its cost.* **And the agreement theorem:** $E(X) = 0$ if and only if the languages agree — six indexes, three operators, no exception (register 1176).

The book's own indexes

What they are. The book at chapter resolution ($E = 578$) and at part resolution ($E = 0$); the register ($E = 6$ at 68.8% density); the reference index (38 cells, $E = 0$, and not a tree); the term index at the back of the main volume (57 terms over 44 specialisation relations, 311 locations, $E = 0$ as a down-set — regenerated from the text at this build).

What only they contribute. *The method self-applied.* **This is the only place where $E > 0$ is a theorem about SHAPE rather than a gap in collection** — the book's chapters do not close because chapters are not a coordinate system, and saying so with a number is the point.

Λ_3 — the three-body index

What it is. Five strata on $\mathcal{M}_{\{E,L\}}$ over three unspecified masses; $E = 0$.

Where it comes from. The classes are Chazy 1922; disjointness is Saari 1971/73 and Painlevé for $n = 3$; the assembly into one index with a certificate is this work's (register 1713).

What only it contributes. *A complete index whose completeness IS the impossibility theorem.* $E = 0$ read as Poincaré: by §25.6 a complete index with no time column predicts nothing, and that is Brudno's rate on the chaotic stratum, not a failure of the method (register 1714, 1724).

The one-line summary

index	the one thing only it gives
Λ	the transfer — cells that are moves
the tower	the price of an axis, measured one at a time
the periodic table	a control the author did not build
Janet	E is coordinate-relative, on one subject
the nuclide chart	a defect whose every cell is nameable physics
the calendar	a subject with no physics in it
box and chessboard	the floor and ceiling of what $E = 0$ means
the EM quotient	quotient versus extension
Λ_{spectra}	values, so the metric defect has something to measure
the violation index	reasoning about an index without holding it
Λ_{phys}	where the work would break first
the languages	translation as re-coordinatisation
the book's own	$E > 0$ as a theorem about shape
Λ_3	a complete index whose completeness is the impossibility theorem — $E = 0$ read as Poincaré

Λ and Λ_3 are the only ones built from physics. The rest are built from Λ , from the act of building Λ , or from drawing the same subject another way — and that is the claim: an index is a coordinate system, and the coordinates come from the subject or from nowhere.

IX · THE INDEXES BUILT IN THE LÖWDIN WORK

Registers 1249–1429. Fifteen indexes: fourteen closed — Λ_{ladder} now among them — and Λ_{spectra} closing at the limit with eleven named cells. The fourteen closed are stated below; Λ_{spectra} is stated in Part VIII. Each is stated the same way: what it indexes, its coordinates, the axis order that closes it, what it refuses, and the one thing it contributes.

Λ_{law} — the laws of the channel equation

What it is. Seven laws on (law, carrier). $E = 0$ at carrier order $u < p < \ell - \ell_{\text{core}} < Z - T$, and 288 of 120,960 orderings reach it.

What only it contributes. *The carrier.* The order is monotone in how LOCAL the variable is — u is the whole atom, p counts one ℓ 's shells, $\ell - \ell_{\text{core}}$ compares two angular momenta, $Z - T$ is one distance to one threshold. **The index says which law runs in which variable, and fitting a law in the wrong carrier is what took the ℓ -spread from a spurious r^2 0.868 to a real 0.356.**

Λ_{const} — the constants

What it is. Fourteen constants on (role, carrier), role order **exponent** < **centre** < **width** < **scale**. $E = 0$ at 2 of 576 orderings.

What only it contributes. *That standing cannot be a coordinate.* With attributed/derived/measured/fitted on an axis the index cannot close at any ordering; without it, it closes at once. **An index whose coordinates mix the object with the observer cannot close, because the observer's axis has no order the object respects.** The same fault recurs as origin in Λ_{var} , kind in Λ_{phys} and state in Λ_{ladder} .

And the role axis orders by how much physics a number has absorbed. Every CENTRE in the system is a small integer or half-integer — 2, 2, 2.5, -1.5, 5.4 — and no SCALE is.

Λ_{var} — the variables

What it is. Twelve variables on (body, role), body order **nucleus** < **core** < **core+rydberg** < **nucleus+core** < **rydberg**. $E = 0$.

What only it contributes. *That there is no nucleus + rydberg cell.* Not one variable relates them directly: every quantity connecting the Rydberg electron to the nucleus — c , u — is a nucleus–core quantity that the Rydberg electron then reads. **That is the screening statement appearing as a structural fact of the index rather than a modelling choice, and it is the three-body factorisation: nucleus–core gives c and u , core–Rydberg gives p , n_0 and T , and the Rydberg electron's own ℓ closes it.**

And the singleton rule. Λ_{var} passes $\mathcal{R}$ with both δ and n^* present, because both land in the same cell — **only a second criterion sees it. An index is closed when its output class is a singleton; two outputs mean the observer is still choosing which to read.**

Λ_{ryd} — the Rydberg series

What it is. Five cells on (Ritz order, ℓ , sign). $E = 0$ once δ_2 is reclassified as INTERMEDIATE — it is fitted from the series, never measured.

What only it contributes. *The n -dependence Λ_{spectra} discards.* A channel's δ is not one number but a convergent sequence — Cd I's ns series gives 3.7171, 3.6835, 3.6719, 3.6665, 3.6637, 3.6621, 3.6610. Λ_{spectra} holds δ_0 ; Λ_{ryd} holds δ_2 and δ_4 .

And the sign of δ_2 is penetration, not ℓ : 0 of 3 positive at $p = 0$, 2 of 2 at $p = 5$. **Seaton's ratio is valid where $p = 0$ at 1.15 ± 0.21 and UNDEFINED where $p \geq 1$** — a domain, not a failure.

Λ_{charge} — the roles of the charge

What it is. Nine occurrences on (role, carrier, sign, regime), nine distinct cells, $E = 0$, and dropping any single role also gives 0.

What only it contributes. *That c is three coordinates in one symbol.* The charge appears inside $u = \ln(N_e/c^{(2/3)})$, as the base of $c^{(-x)}$, and inside the exponent x itself. **Every degeneracy found was between a charge term and something else, and this index says why: one symbol occupying three positions, and any fit letting two of them float will trade them.**

A warning the index also carries: declaring the three roles as coordinates in Λ_{spectra} raises E from 1929 to 13,605. **A closed sub-index says its roles are well-posed; it says nothing about whether the parent index wants them.**

Λ_{cross} — the crossing values

What it is. The value at which an ordering flips, $a_{\text{cross}} = \Delta n(\sqrt{p_g} + \sqrt{p_r})/(p_g - p_r)$ — nineteen distinct surds across the table, including $1/\sqrt{3}$, $1/\sqrt{2}$, 1, $1/(\sqrt{3}-1)$, $1+1/\sqrt{2}$ and $1/(\sqrt{2}-1)$. $E = 0$ on $(\Delta n, \ell_g)$.

What only it contributes. *An index with no statistics language.* Order, algebra (closed in $\mathbb{Q}(\sqrt{p})$ for $p \leq 7$) and geometry all speak of it; **analysis and statistics do not.** A quantity that is ordinal, algebraic and geometric but not analytic is an INTEGER OBJECT: it cannot be fitted and needs no fitting.

And it is the one index whose unfixable cell is fixed anyway — a node count needs no measurement. **Nine of eleven unfixable cells across the work are expectation values $\langle \Psi | \hat{O} | \Psi \rangle$; Λ_{cross} is the exception, and it is the only index with no statistics language. The two facts are the same fact.**

Λ_{descent} — the charge dependence

What it is. $a(c)$ on (c, regime) . $E = 0$.

What only it contributes. *The only place a fit belongs.* Order and analysis both speak of it, information says c adds everything, and statistics answers yes. **Every fit made was applied to the WHOLE of a — the crossing and the descent together — and that is why the constants kept absorbing each other: half the object has no analysis language, and the fitted parameters were competing to represent a set of surds.**

Λ_{phys} — the parameters, rebuilt

What it is. Twenty-one parameters of real atoms on (source, domain), $E = 0$, at source order **standard** < **mathematics** < **literature** < **this work** and domain order **universal** < **all elements** < **a region** < **one species**.

What only it contributes. *That no parameter of this work is universal.* The closing grid is a staircase and the diagonal is the whole content: CODATA constants are universal, this work's numbers hold on a region or all elements and never universally. **A pooled fit across regions asserts a universal parameter, which the closed index says does not exist — that is a structural prohibition, not a stylistic preference, and it is what the closure of the domain axis enforces.**

Rebuilt on three rulings: kind is the same axis as source; arity is a property of the book, not the parameter; and parameters that are artefacts of the METHOD are not physics of real atoms.

Λ_{amp} — the electron–electron term

What it is. The Slater integrals: for each subshell pair, F^k for k even up to $2\min(\ell, \ell')$ and G^k for $k \equiv \ell + \ell' \pmod{2}$ from $|\ell - \ell'|$ to $\ell + \ell'$. **Twenty cells, $E = 0$, a perfect triangle $0 \leq i \leq \ell$.**

What only it contributes. *That the rank is not a free coordinate.* Indexed on the multipole rank k the index gives $E = 1$ and the single defect is F^1 — **a term parity forbids**. Reindexed on POSITION WITHIN SEQUENCE it closes. **The same fault as breadth in Λ_{chem} : an axis whose values are constrained by another axis.**

And its structure explains the Slater result exactly: at s/s the exchange G^0 IS the direct F^0 , so dropping exchange costs nothing; at f/f it discards four independent quantities. **The count of lost integrals is the count of failures — 5 of 5 s-block brackets threaded, 0 of 1 f.**

Λ_{PCA} — physics $\oplus$ charge $\oplus$ amplitude, merged

What it is. Nine cells on (source, domain) with the domain refined by the charge regime: **universal < all elements < low < neutral < hydrogenic < one species.** $E = 0$.

What only it contributes. *The per-atom calibration.* A parameter's domain reads as a SET OF ATOMS, so for any (Z, c) the index states which parameters apply to it. **And the closing order puts low — charge 2 — BEFORE the neutral, which the ladders independently confirm: $c = 2$ is the only charge with two-sided brackets, the narrowest domain and therefore the most particular.**

A correction recorded here: merging Λ_{phys} and Λ_{charge} on identified axis NAMES raised E from 6 to 11 and I concluded they were not one object. That was against a malformed Λ_{phys} . **Merged on domain $\equiv$ regime alone, after the rebuild, they close.**

Λ_{chem} — the chemical properties

What it is. Forty-two chemical properties of a species on (kind, seat, PCA dependency). $E = 0$ on fourteen cells over the subvalence and valence shells.

- **kind:** count · symmetry · size · energy · rate
- **seat:** the nucleus · the core · the subvalence shell · the valence shell · **the aggregate**
- **PCA:** which of physics, charge or amplitude the property supplies

What only it contributes. *That twelve of the forty-two are properties of MATTER IN BULK and not of an isolated atom* — density, melting point, hardness, crystal structure, conductivity, colour of the metal, smell, taste, metallic character, reactivity. **With eleven hand-picked properties the index could not say this; it appeared the moment every property was demanded.**

And its closure states PCA's own boundary. E climbs $0 \rightarrow 3 \rightarrow 8 \rightarrow 11 \rightarrow 19$ as the core, the nucleus and the aggregate are added. **PCA's domain is the subvalence and valence shells, and the index declares it by degrading monotonically outside it** — as Seaton's ratio is valid at $p = 0$ and undefined beyond.

The last defect cell, found by exhaustive reordering. All 1,440 orderings give minimum $E = 1$ and none reaches zero: the cell is **symmetry $\times$ the valence shell $\times$ a charge role**. The fill is the ground TERM, which changes with charge — $N_e = 20$ gives $^1S_0 \rightarrow ^3D_1 \rightarrow ^3F_2 \rightarrow ^3F_2$ and $N_e = 38$ the identical sequence. **Every ladder table carried the term symbol; the configurations were recorded and the terms were not.**

Λ_t — the corridor position

What it is. Twelve cells on (ℓ, n) holding $t = (a - L)/(U - L)$, where a sits in its corridor. **$E = 3$.**

ℓ n	2	3	4	5	6	7	limit
s	0.268	0.335	0.266	0.218	0.257	0.195	—
p	·	1.120	1.049	1.022	1.002	·	1.0000
d	·	·	1.819	1.433	·	·	1.7321

What only it contributes. *That the ℓ axis carries the LIMIT and the n axis the APPROACH.* t converges to $\sqrt{(\ell(\ell+1)/2)}$, the centrifugal term — 6p sits 0.2% above it — and the gap falls **$0.120 \rightarrow 0.049 \rightarrow 0.022 \rightarrow 0.002$** along n .

And q is not a coordinate. With the Pauli fraction in the radicand, t barely moves across a subshell: 6p gives 1.0237, 1.0417, 1.0009, 1.0036, 0.9946, 0.9940 across all six occupancies.

Its three defect cells are 7p, 6d and 7d — real absences, each a subshell whose corridor is one-sided or whose ionisation energy is not held.

Λ_{ladder} — the ladders themselves

What it is. The walks, on (seat, kind, Zcross). **FOURTEEN ladders in three families — six species, six state, two X-ray — nine cells, $E = 0$. IT CLOSES.** Registers 1356, 1374, 1375, 1384, 1427.

Two relations, three ladders each — $Z = N_e + c - 1$ on Λ , and $A = Z + N$ on the nuclide chart. Only two of three are independent either side, so no ladder varies Z alone: forbidden by arithmetic, not by lack of data.

family	ladder	fixes	reaches	held
species	isoelectronic	N_e	valence	11 of 11
species	the walk	c	valence	106 of 106
species	ionisation	Z	subvalence	15 of 108
species	isotopic	Z, N_e, c	nucleus	1 rung traced
species	isotonic	N	nucleus	none — NEW
species	isobaric	A	nucleus	none — NEW
state	the Rydberg series	the species	valence	every δ held
state	the ℓ -ladder	species, n	valence	62 pairs
state	the term ladder	species, cfg	valence	66 pairs
state	the outer- j ladder	species, cfg	valence	the j -splitting (register 754)
state	the parent-term	the species	core	12 of 15
state	the isomeric	Z, N, e^-	nucleus	none — NEW

What only it contributes. *That a coordinate individuating the cells is a KEY, not an axis.* At four ladders fixes was injective by construction — a ladder IS named by what it holds fixed — so one cell per row, and **E = 0 was FORCED: 2% of admissible arrangements could refuse.** At six, ionisation and isotopic both fix Z, and the zero becomes earnable. That is the dual of the defining letter (§21.6.1), where a single-valued coordinate contributes no envelope; this is the other end of the same degeneracy.

IT CLOSES, and both blocking reasons are resolved. The E = 1 defect stood at (**subvalence, counting, across elements**) — Moseley’s ladder, in a direction ionisation does not go. That defect is what demanded Λ_{xray} , and Λ_{xray} filled it: Moseley enters at subvalence as counting/across, the $K\alpha$ doublet as coupling/within. **Only the subvalence seat closes** — nucleus gives 1, core 2, valence 1 — and the seat is settled from OUTSIDE the closure rather than by it, since Λ_{PCA} excluded the core independently, on different coordinates, before this was computed. The physics agrees: a $K\alpha$ line is not a property of the 1s shell but a transition *between* shells, so the seat is where the transition spans, not where the hole sits. **Contingency: 90% of comparable nine-cell sets refuse, so the zero is earned** (R 1427).

And the five shared cells are four findings and one separation (R 1428). Two pairs are ONE OBJECT and the index is right to merge them — isotonic fixes N and isobaric fixes A with $A = Z + N$; isoelectronic fixes N_e and the walk fixes c with $Z = N_e + c - 1$. That is *charge is one symbol in three positions*, one level up. Two are separated by axes already in the index but not among the three that close: isotopic is species where the isomeric is state; the Rydberg series fixes the species where the ℓ -ladder fixes species and n. **The fifth was the real ambiguity** — the term ladder and the outer-j ladder — and they are five tower stages apart, nested rather than parallel: the term ladder moves 2S, Λ_8 ’s own letter, while the outer-j ladder moves 2J, which arrives only at Λ_{13} . Adding the tower stage as a fourth axis separates that pair and only that pair, but costs closure, E going $0 \rightarrow 1$. **So the stage is the discriminator, not an axis** (R 1429).

Λ_{xray} — the inner shell

What it is. The dipole-allowed inner-shell transitions, on $(\Delta n, \Delta \ell, j_{\text{type_hole}})$. **33 lines through the N shell $\rightarrow$ 9 cells, E = 0.** Register 1378.

Where it comes from. Built with no new data. **Λ ’s cell IS a transition** — source subshell into target subshell — and its constraints impose no order between n and e, so a downward transition was already inside its alphabet; only the caps excluded it. The electric/magnetic multipole selection rules (§12.11.8) give the dipole rule, the recoupling bound (§12.11.1) the j triangle. The six K-shell lines emerge as the classical set unadjusted: K-L2 and K-L3 are $K\alpha_2$ and $K\alpha_1$, K-M2 and K-M3 are $K\beta_3$ and $K\beta_1$, K-N2 and K-N3 are $K\beta_2$. Energies are NIST SRD 128, Deslattes *et al.*, *Rev. Mod. Phys.* **75** (2003) 35–99.

What only it contributes. *That thirty-three named lines are nine transition types.* Seventy-two of 165 three-coordinate systems reach E = 0 and **every one collapses to exactly nine cells** — $K\alpha$, $L\alpha$ and $M\alpha$ are one type read at three depths, which Siegbahn notation hides.

And it settles what Λ_{cross} ’s silences mean. The two close on the same shape — differences plus one endpoint — and the cypher separates them:

language	Λ_{cross}	Λ_{xray}
order	speaks	speaks, E = 0
geometry	speaks	speaks
statistics	silent	speaks — pairwise marginals recover all 9
analysis	silent	speaks — Moseley, $R^2 = 0.998$

Λ_{cross} remains the only integer object: cells AND values fixed by arithmetic. Λ_{xray} has integer cells and measured values. Registers 1379, 1380, 1385.

The one-line summary, extended

index	the one thing only it gives
Λ_{law}	the carrier — which variable a law runs in
Λ_{const}	that standing cannot be a coordinate
Λ_{var}	no nucleus + rydberg cell — the three-body factorisation
Λ_{ryd}	the n-dependence Λ_{spectra} discards
Λ_{charge}	c is three coordinates in one symbol
Λ_{cross}	an index with no statistics language
Λ_{descent}	the only place a fit belongs
Λ_{phys}	no parameter of this work is universal
Λ_{amp}	the rank is not a free coordinate
Λ_{PCA}	the per-atom calibration
Λ_{chem}	twelve of forty-two properties are not an atom's at all
Λ_{t}	ℓ carries the limit, n carries the approach
Λ_{ladder}	that a coordinate individuating the cells is a key, not an axis
Λ_{xray}	thirty-three named lines are nine transition types
Λ_{chain}	Λ 's input column as a derived object — the order from the equation, one constant
Λ_{cinf}	that the periodic table is not a solution of the non-relativistic equation
Λ_{V5}	every contested competition widens under correlation
Λ_{j120}	Λ 's falsification frontier, twelve rows wide
Λ_3	masses enter through six numbers; structure is mass-free — the three-body factorisation, closed

Fourteen close. Λ_{xray} is new and closes at nine transition types. **Λ_{ladder} now closes too** — fourteen ladders, nine cells, $E = 0$, once the X-ray pair is seated at subvalence (R 1427), its five shared cells resolving as four findings and one separation (R 1428–1429).

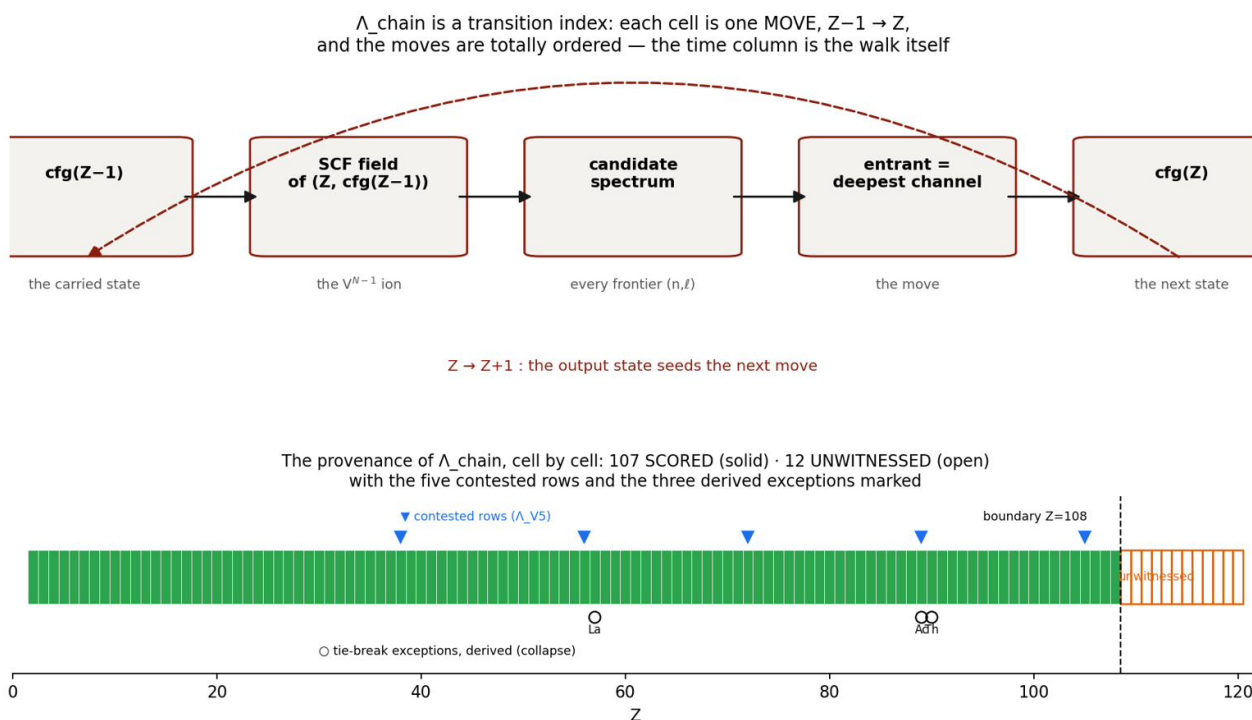
Λ_{spectra} closes at the LIMIT — not for want of the principal number, as register 1341 supposed. The index runs to the last available species and stops: 98 cells, $E = 58$, of which 38 need more electrons than any atom has. **The remaining twenty are within the limit, and eleven of those are the Madelung exceptions, each nameable** (R 1395, corrected at R 1426).

X · THE INDEXES BUILT IN THE LÖWDIN SOLUTION

Registers 1701–1712; Chapter 35. Four indexes the solution builds beside Λ — what each holds, whether it closes, whether it carries time, and what role it plays for Λ . The template differs from Part IX's by design: these are transition indexes, and closure and time are the questions they exist to answer.

Λ_{chain} — the derived filling index

What it holds. 119 rows, one per element $Z = 2\text{--}120$: entrant channel, entrant depth, margin, and the full frontier candidate spectrum of the V^{N-1} walk; provenance per row (SCORED 107, UNWITNESSED 12).



Does it close? As an index of rows, trivially — one cell per Z , none refused. The substantive closure is different in kind and stronger: **the chain closes on itself**. Each row is built from the previous row's derived output, so a single wrong cell breaks every cell after it, and 107 of 107 is therefore one score, not 107 (register 1701–1702).

Does it carry time? Yes, in this compendium's exact sense: **its cells are moves** — the step $Z-1 \rightarrow Z$, adding one electron to a channel — so it carries a time column, standing beside Λ as a second transition index in the collection, and the first whose moves are **totally ordered**. Chapter 34 proved the table needs a carried state; Λ_{chain} is that state, carried by the equation.

Role for Λ . It supplies, from first principles, the one thing Chapter 6 proved Λ cannot supply for itself: the aufbau ordering. Λ 's **input column is now a derived object**. Every configuration Λ indexes is generated by Λ_{chain} with one constant, and the two agree everywhere Λ has an element.

Λ_{cinf} — the counterfactual twin

What it holds. The identical 107-row walk at $c \rightarrow \infty$.

Does it close? As Λ_{chain} does, on itself.

Does it carry time? Yes — the same move-cells as Λ_{chain} , in the same total order.

Role for Λ . None as data, everything as contrast: **eleven of its entrants differ from Λ_{chain} 's, and all eleven are wrong against nature.** It exists to state that Λ 's subject is relativistic, and it is quarantined from every other use (register 1706).

Λ_{V5} — the contested-row closure

What it holds. Five rows — $Z = 38, 56, 72, 89, 105$ — each carrying the complete second-order correlation differential over the mean-field margin, with declared envelopes.

Does it close? Yes, by exhaustion: the five rows are *all* the rows whose margins are small enough to be at risk, and the criterion selecting them is stated, not curated.

Does it carry time? No. It is a still photograph of the walk's five closest calls.

Role for Λ . It is the warrant that Λ_{chain} 's mean field carries the exact equation's order: **every contested competition widens under correlation** (register 1705). Without this index the derivation would be conditional in a place it is now checked.

Λ_{j120} — the unwitnessed extension

What it holds. Twelve rows, $Z = 109\text{--}120$: entrant, depth, margin, and the spin-orbit worst case each margin clears.

Does it close? Its cells do; its **values are unwitnessed** — the grade this book already owns for a bound the world has not yet supplied. Nothing in it is defected for want of trying.

Does it carry time? Yes — move-cells, Λ_{chain} 's own order continued past the last measurement.

Role for Λ . It is Λ 's falsification frontier. The day a superheavy ground configuration is measured, exactly one row here either scores or fails, and either outcome is information the whole structure inherits (register 1712).

One line for the map: Λ_{chain} stands to Λ as derivation stands to arrangement. Λ holds the configurations; Λ_{chain} says why they are the ones held — and says it from the equation, with one number, carrying the state that closure alone could never carry.

XI · TRANSITIONS, THE COMPLETE TABLE

Every section of the source, not only the ones the book leans on. *Transitions* v3.0 is the origin of the closure operator R, the excess E, and the two failure modes on which Part IV of this compendium and Chapter 12 of the main volume both rest. Thirty of its sections are load-bearing here, and the main volume's Appendix G sets those thirty out with the objects that stand on each. This section is the other half of that address: **all seventy-seven sections, in order, with what each states and whether this work rests on it.**

Why the forty-seven are printed at all. An index of indices that lists only what a work used tells a reader nothing about what it declined. The forty-seven are the paper's own ordering arguments, its withdrawals, its axis index and its open join — several of them corrections to sections this book *does* use, and one of them, §10.2, a correction to §2.8 that reverses a claim about the measure. A reader who wants to know whether a result of this book was picked from a larger field or was the only thing on offer can only answer that from the complete list.

The count. Seventy-seven sections; thirty relied on, forty-seven not. The paper remains the origin and is named as such throughout; this table and Appendix G are the address.

§	what it states	in this work
0.1	Five working protocols, the first of which is commit before looking: a position is written down before any retrieval, and the retrieval scores it.	not used
0.2	The Admission Law. An entry's grade bounds the operations it may enter; a derived entry carries the minimum grade of its inputs, and no derivation raises a grade.	not used
0.3	A biconditional licenses a merge only if it holds at every rung of the graded axis. Merging on a fact true at one point and then grading the axis is the error.	not used
0.4	Operator reliability, measured. Predictions aimed at logical structure scored 4/4 on three consecutive tests; predictions of how a field regards a claim scored 1/4, and that class was barred from entering.	not used
1.1	The definitions: the value sets, the monotone upper envelope ϕ_{ij} of one coordinate against another, $R(X)$ as every point of the observed box respecting all envelopes, and $E(X)$ as its excess.	not used
1.2	$X \subseteq R(X)$, and therefore $E \geq 0$. Extensivity, proved.	rests on it — Appendix G
1.3	R is a closure operator — extensive, monotone, idempotent. The closed sets form a Moore family.	rests on it — Appendix G
1.4	R is the closure of a binary constraint network under monotone binary projections, and $E(X) = 0$ exactly when the network is globally consistent.	rests on it — Appendix G
1.5	$X \subseteq BPC(X) \subseteq R(X)$, so $E > 0$ is in principle ambiguous between genuine inconsistency and envelope coarseness. Computed for every object in the paper, $R = BPC$ exactly and the ambiguity does not arise.	not used
1.6	The two failure modes — order and arity — and the third case: a vocabulary may not be indexable at all.	rests on it — Appendix G
1.7	Six blindnesses of the closure. A derived coordinate cannot repair it: the box grows by that coordinate's value count while \	rests on it — Appendix G
1.8	The closure rule. A tightening preserves $E = 0$ if and only if it binds one coordinate by a monotone function of one other. One-sided as stated; the operator is two-sided (§2.15.2; register 402).	rests on it — Appendix G
1.9	E requires density, asymmetrically. $E = 0$ is informative at any density; $E > 0$ at low density measures sparsity and nothing else.	rests on it — Appendix G
1.10	The two named theorems covering Λ — Freuder 1982 and Montanari 1974. Neither reaches either violation index.	rests on it — Appendix G
2.1	The tower closes at every level for every statistics order. Parastatistics of order m gives	rests on it — Appendix G

§	what it states	in this work
	capacity $m(4\ell+2)$; $m = 1$ reproduces the canonical tower exactly.	
2.1b	Closure is not scheme-contingent. Rebuilt in LS, LK, JK and jj: the coupling scheme is not a coordinate.	rests on it — Appendix G
2.2	Λ_9 's constraint graph is a tree of nine nodes and eight edges, so Freuder applies. The admissible Pauli cut gains a cycle and loses the guarantee. Λ_9 is the last tree level.	rests on it — Appendix G
2.3	All six density values reproduce once three non-obvious exact sets are used — the spin set of $f^\wedge g$ by microstate enumeration at axis 9, terms genuinely new at occupancy v at axis 10, and the J values of terms of ℓ^k at axis 11.	not used
2.4	Closure and exactness are incompatible. The exact coupling triangle is ternary, outside the closure-preserving class: Λ_{12} falls $70,905 \rightarrow 22,275$ and E goes $0 \rightarrow 35,570$.	rests on it — Appendix G
2.5	Four monotone variants of the ℓ -bound close and two non-monotone variants do not. The decisive pair, $\ell \leq \lfloor n/2 \rfloor$ and $\ell \leq \lfloor n-3 \rfloor$, have identical cardinality at 32,535 cells and opposite outcomes: monotonicity is the discriminant, not cardinality.	not used
2.6	The Standard-Model Extension lifts the Coulomb accidental degeneracy but moves no quantum number, because $\ell \leq n-1$ follows from node counting. The lattice does not individuate universes by Lorentz structure.	not used
2.7	Transit admissibility is directional. Outward is unobstructed; inward is not, because a cell with occupancy beyond our Pauli bound fails the one-dimensional index before any question of joint structure arises.	not used
2.8	Three things locate a transition and the index certifies only the first. The lattice says what exists; the order says what is near, and only 24.2% of the 475,800 pairs of Λ_8 cells are ordered by componentwise domination; the measure says what it costs, and it is external.	not used
3.1	The eighteen-column table has $E = 36$, and re-indexing period as $n + \ell$ is not what repairs it — neither is contiguity. ℓ is fully determined by group: s at 1–2, d at 3–12, p at 13–18.	not used
3.2	Of the 36, twenty-five — $1d$, $1p$ and $2d$ — could never hold an element, being forbidden by $\ell \leq n-1$: the footprint of a law the coordinate system has no axis for. The remaining eleven are $3d$, deferred past $4s$ by the Madelung order, and period 1 group 2.	not used
3.3	The periodic table is not an analogy for the violation index; it is the other failure mode, and having both is what makes the distinction visible. Ordering failures are repairable because re-ordering acts on one relation at a time.	not used

§	what it states	in this work
4.1	Λ_9 is the transit level. 1,169 of 1,654 cells are composable — the first level at which a transition has a defined endpoint, and the last that is a tree.	rests on it — Appendix G
4.2	The composition graph is the line digraph of a quiver on 33 atomic states, with 27,027 edges. A loop sits at every vertex, so return is available in one step.	rests on it — Appendix G
4.3	Girth exactly 4. Unit-step adjacency coincides with the covering relation at 6,658 edges either way, so the lattice is gap-free.	rests on it — Appendix G
4.4	Molecular transit. Feasibility decays geometrically at 2.51 per atom; the Helly number is at least 5 and at most 144.	rests on it — Appendix G
5.1	The coordinates of the violation index, and the position this work occupies in it — two components fixed by measurement rather than choice.	rests on it — Appendix G
5.2	Chronology violation requires the causal ladder, non-unitarity at the Lindblad rung and arbitrarily small ANEC violation — and not signalling, cloning, nonlinearity or loss of microcausality, each of which appears under Deutsch's prescription and not otherwise.	not used
5.3	2,370 cells, $E = 30$, all of it genuine global-consistency failure. The excess is exactly 1×30 and the core is a single cell.	rests on it — Appendix G
5.4	The failing triple is the unique minimal support. Arity exactly 3, one generator; no single literature pays for it.	rests on it — Appendix G
5.5	The repair space is closed, and both directions fail for opposite reasons: adding fails because the box inflates, merging fails because the envelopes coarsen. Two lines cover all six repairs.	not used
5.6	Objects are thresholds. Of the 1,146 cells permitting a macroscopic wormhole, 1,116 pay with a preferred frame, 714 with non-unitarity and 756 with signalling — and none pays with nothing.	not used
5.7	The defect does not scale. The core is one cell at every resolution, and the multiplicity moves non-arithmetically.	rests on it — Appendix G
6.1	Of the nine coordinates, five are fully conflated, two partially, and two are clean.	not used
6.2	Every conflation merges an inert rung with a potent one, and that is not coincidence: the index was graded from papers about what breaks, and the papers that matter for atomic structure are about what holds. Two different literatures.	not used
6.3	None of the five was findable from inside. All surfaced from a specific question about a named theorem; inspecting the coordinate list finds none of them.	not used

§	what it states	in this work
6.4	Audit 22. For every coordinate whose term appears in a cited theorem, state what the theorem means by it, state what the rung means, and record match, conflation or unchecked. Its output is a debt, not a pass.	not used
6.5	The corrected alphabet — what becomes expressible once the conflated letters are split.	rests on it — Appendix G
7.1	Every structural result survives at fifteen letters: E = 816, one cell, arity 3.	rests on it — Appendix G
7.2	Recomputed under the documented nine-letter conventions the linearisation figure reproduces exactly at 7,734; the slide figure does not, and the earlier 8,856 is refuted outright, the operation acting on a three-coordinate object whose box is at most 40.	not used
7.3	The frontier formulas in closed form, d(c) and s(c), verified on 1,500 sampled cells with zero distance mismatches.	not used
7.4	The spin-statistics hypotheses have three letters and no proxies. 900 of 18,072 cells — 5.0% — have the Pauli lattice guaranteed, and we are among them; for the rest parastatistics becomes available and Λ_8 can be 1,600 rather than 976.	not used
8.1	A charge relation has three parts, not two: trigger, what incurs the charge; currency, what may pay; and jurisdiction, where the charger has standing to collect.	not used
8.2	A jurisdicted forcing and an unjurisdicted disjunction are the same object. A theorem holding only under a scope condition is inherently ternary, and no further physics removes the scope.	rests on it — Appendix G
8.3	Buniy’s jurisdiction is necessary, and the counterexample is the ghost condensate of Creminelli, Luty, Nicolis and Senatore (2006). It escapes through higher-derivative structure, not Lorentz violation: the Lagrangian is manifestly Lorentz-invariant and the vacuum breaks the symmetry.	not used
8.4	The vocabularies. A locally covariant QFT is a functor with exactly four parts, which replaces the seven-vocabulary partition.	rests on it — Appendix G
8.5	Retained as history, not current analysis. Built on the seven-vocabulary partition that §8.4 derives away and §12.3 withdraws, and kept because §8.6’s correction is only legible against what it corrects.	not used
8.6	An earlier draft placed the exclusion of macroscopic wormholes in a component the law index cannot reach, and made that the reason six repairs failed. Rebuilt on the four derived vocabularies, that is wrong: the system is connected and the severance was an artefact.	not used
8.6b	Graham–Olum is listed by the charger index as firing and should be conditional. A conjecture	not used

§	what it states	in this work
	with a sufficiency proof is a third category the index does not have.	
8.7	Retained as history. No configuration is a connected tree; Hartman must drop V2 rather than V3, and reducing Wall backfires because its V1–V5 edge is coupling’s only connection.	not used
8.8	Checked twice, including through an adjacent literature: the general ANEC-from-causality argument is flat-space, and curved-space results exist only with a V2 condition retained.	not used
9.1	Three measures, and the third did not exist. E measures what an index cannot carry; the vocabulary debt measures what it has not checked it carries correctly, 32 of 48; nothing measured what was never named at all, and seven law-classes appear in cited sources with no letter.	not used
9.2	The axis index itself — nine cells in a box of 5,184.	not used
9.3	Pair-completion. The nine leave four value-directions unoccupied, and every hand-found axis outside the box occupies a pair — never one direction, never three.	not used
10.1	Every vocabulary index closes; the charger index does not. The defect sits in V1, on the V1–V3 edge, and in the partition itself.	rests on it — Appendix G
10.2	§2.8 is corrected. A faithful measure exists and compresses 976 cells to eighteen values, because any strictly monotone function of the coordinates is automatically faithful.	not used
10.3	Λ audited. The conflation rates of Λ and the violation index are proportionally indistinguishable; the causes differ, V1 conflating because two literatures used one word.	not used
10.4	Six independent curved-space routes to the ANEC. Five require a Killing field or a horizon generated by one; the sixth yields a weighted bound instead.	rests on it — Appendix G
10.4b	The one open cell. Following §10.4’s target down gives a single surface class and a single condition, and the target is a boundary point rather than a region; an isolated horizon has $\Theta = 0$ exactly.	not used
10.4c	C1, computed. The presymplectic potential on a null surface contains no transverse derivative, so Ω is block diagonal and the algebra factorises over generators.	rests on it — Appendix G
10.4d	C2, the open condition — whether the state respects the decomposition. Independent of the Hadamard condition, which says nothing at finite transverse separation.	rests on it — Appendix G
10.4e	Half-sided modular inclusion is the converter from algebra to geometry: Borchers and Wiesbrock return the affine group acting on a line.	rests on it — Appendix G
10.5		rests on it — Appendix G

§	what it states	in this work
	Status is one bit per vocabulary and the partition forces one-hot — every term is physical in exactly one vocabulary, or in none.	
10.6	The chain, primary-sourced, with a fifth caveat on T0: Reeh–Schlieder is a theorem in Minkowski space, proved from analyticity, the spectrum condition and the action of the Poincaré group, and in curved spacetime it is a property established for free massive fields.	not used
10.7	Five instances of one shape, the last stated as physics rather than bookkeeping: gauge redundancy is the bookkeeping needed to describe subsystems relationally in a gauge-invariant system.	not used
12.1	The promotion ledger. Everything promoted is about the method and everything held is either a general claim or a physical number — the instrument is better characterised than anything it measured.	not used
12.2	Twelve claims withdrawn with their causes, among them defect 60, an unbounded Helly number, projection-covariance as a scaling law, and the defect growing with jurisdiction count.	not used
12.3	The withdrawals post-dating the previous draft, several of which correct claims that draft asserts.	not used
12.4	The numeric debt. Of six claims carried with no dataset behind them, five clear on recomputation and one is corrected — refuted by a bound, the slide operation acting on a three-coordinate object whose defect cannot exceed 40.	not used
12.5	Audit 23 is the only external validation in the construction and it passes: ten published LS term tables reproduced exactly, zero microstate discrepancies against $C(4\ell+2, k)$.	not used
12.6	Thirty instances of one class — a conclusion drawn from a comparison that was not licensed. The arithmetic was correct in every case; the warrant was not.	not used
12.7	Four of the seven propositions are refuted, two unassertable and one conditional. None closes the index, and with Buny’s jurisdiction established as necessary, P5’s conditionality no longer decides the matter.	not used
13.1	For a product with no linking constraint every cross-envelope is vacuous, so $R(A \times B) = R(A) \times R(B)$. With $E(\Lambda) = 0$ the join of Λ with the violation index gives $E = 976 \times 30 = 29,280$: the defect is extensive.	not used

Reading the third column. *Rests on it* means at least one object of this work — a lemma label, a chapter, or a Register entry — cites that section; Appendix G names which. *Not used* means no object cites it, and carries no judgement on the section: §12.5’s external validation and §13.1’s extensivity result are both sound and both simply outside what this book claims.